

PB2000-104672



Experimental results of motions, hydrodynamic coefficients and wave loads on the 372 catamaran model

Ir. Riaan van't Veer

Report 1129

February 1998

REPRODUCED BY: **NTIS**
U.S. Department of Commerce
National Technical Information Service
Springfield, Virginia 22161

TU Delft

Delft University of Technology

Faculty of Mechanical Engineering and Marine Technology
Ship Hydromechanics Laboratory

Contents

Abstract	2
List of Symbols	3
1 Description of the experiments	5
1.1 Introduction	5
1.2 Experimental set-up	6
1.2.1 The still water resistance tests	7
1.2.2 The heave and pitch motion tests	8
1.2.3 The forced heave and pitch oscillation tests	9
1.2.4 The wave load tests	10
1.3 Test program overview	11
2 Measurements results	15
2.1 The still water resistance tests	15
2.2 The heave an pitch motion tests	21
2.3 The forced heave and pitch oscillation tests	26
2.3.1 Forced heave oscillation tests	27
2.3.2 Forced pitch oscillation tests	27
2.3.3 Restoring coefficients	28
2.4 The wave load tests	46
Acknowledgment	51
References	52

Abstract

This report presents the measurement results of a series of experiments carried out in the towing tank no 1 at Delft University of Technology. The measurements were carried out as part of a PhD project.

The comparison of the measurements with numerical results is presented in the PhD thesis Van 't Veer (1998a). Another series of experiments with the catamaran 372 in oblique waves is presented in the measurement report Van 't Veer (1998b).

List of Symbols

a	-	subscript for amplitude
Δ	[kg]	displacement
ϵ_3	[deg]	heave phase angle
ϵ_5	[deg]	pitch phase angle
η_3	[mm]	heave motion
η_5	[rad]	pitch motion
λ	[m]	wave length
ρ	[kg/m ³]	water density (1000 kg/m ³)
θ	[rad]	pitch angle
ω	[rad/sec]	wave frequency
ω_e	[rad/sec]	wave encounter frequency
ζ	[mm]	wave height
A_{ij}	[]	added mass value in the i -direction due to a force in the j -direction
B_{ij}	[]	fluid damping value in the i -direction due to a force in the j -direction
B	[m]	overall vessel beam
C_{ij}	[]	restoring coefficient in the i -direction due to a force in the j -direction
C_B	[-]	block coefficient
F_3	[N]	Vertical force (in the z -direction)
F_5	[Nm]	Moment around the y -axis
F_n	[-]	Froude number; $F_n = U/\sqrt{Lg}$
H	[m]	distance between the centre lines of the two hulls
I_{55}	[kgm ²]	moment of inertia, around y -axis
KG	[m]	distance of centre of gravity above keel
LCG	[m]	longitudinal centre of gravity, measured from AP
L_{0A}	[m]	overall ship length
L	[m]	ship length between perpendiculars
M	[kg]	mass

R	[N]	overall ship resistance
T	[m]	ship draught
T_{AP}	[m]	draught at aft perpendicular
T_{FP}	[m]	draught at fore perpendicular
U	[m/sec]	steady forward speed of the vessel
b	[m]	breadth of the demihull
g	[m/sec ²]	gravitational acceleration; 9.81 m/sec ²
k	[rad/m]	wave number; $k = \omega^2/g$
k_{yy}	[m]	pitch radius of gyration
x	[m]	x-coordinate
y	[m]	y-coordinate
y_w	[m]	local beam of vessel, centre line to waterline
z	[m]	z-coordinate
z	[mm]	heave motion

1 Description of the experiments

1.1 Introduction

To obtain insight in the behaviour of fast catamaran vessels in waves, a series of experiments have been carried out. The test program consisted of:

- Resistance measurements, including trim and sinkage registration,
 $F_n = 0.18$ to $F_n 0.75$
- Steady wave pattern measurements using a longitudinal wave cut,
 $F_n = 0.30$ to $F_n 0.75$
- Steady hull wave pattern visualisation via photo's,
 $F_n = 0.30$ and 0.45
- Heave and Pitch motions registration in head waves,
 $F_n = 0.30, 0.45, 0.60$ and 0.75
- Heave and Pitch oscillation tests, zero trim and sinkage,
 $F_n = 0.30, 0.45, 0.60$ and 0.75
- Heave and Pitch oscillation tests, trim and sinkage correction,
 $F_n = 0.75$
- Wave force measurements, head waves, zero trim and sinkage,
 $F_n = 0.30, 0.45, 0.60$ and 0.75

The catamaran used in the experiments was designed at Delft University. The ratio of several overall dimensions was chosen from realistic values found in the literature of recently build catamarans.

The transom was given a zero depth. It is understood that for practical reasons the transom stern of a catamaran has a certain depth, since this is necessary to install the waterjet propulsion system. However, to illuminate the highly nonlinear behaviour of such a transom stern, a stern with zero depth was chosen.

The range of forward speed in which the catamaran model was tested, was selected from $F_n = 0.30$ up to $F_n = 0.75$, which is a realistic operating range for existing catamaran vessels.

1.2 Experimental set-up

Model experiments with the 372 catamaran model were carried out in the towing tank no. 1 of the Delft Ship Hydrodynamic Laboratory. The dimensions of the towing tank are $L \times B \times D = 145 \times 4.2 \times 2.6$ [m]. During the experiments the water depth varied between 2.33 m to 2.40 m, due to the restrictions of the experimental equipment.

The main particulars of the catamaran vessel are given in Table 1.1 and a lines plan of the catamaran is presented in Figure 1.1. At the end of this chapter a list of ordinates is included and a second lines plan is given (large scale).

Main Particulars	Symbol	Value
Length over all	L_{OA}	3.11 m
Length between perpendiculars	L	3.00 m
Beam overall	B	0.94 m
Beam demihull	b	0.24 m
Distance between center of hulls	H	0.70 m
Draught	T	0.15 m
Displacement	Δ	87.07 kg
Draught, AP	T_{AP}	0.15 m
Draught, FP	T_{FP}	0.15 m
Vertical center of gravity	KG	0.34 m
Longitudinal center of gravity	LCG	1.41 m
Pitch radius of gyration	k_{yy}	0.782 m
Moment of inertia for pitch	I_{55}	53.245 kg m ²
Length over beam ratio	L/b	12.5
Length over draught ration	L/T	20.0
Block coefficient	C_B	0.403
Mass during oscillation tests	M	62.00 kg
Moment of inertia, osc. tests	I_{55}	13.454 kg m ²

Table 1.1: Main characteristics of the Catamaran 372 model.

The experimental program was divided in four different test series which are described below.

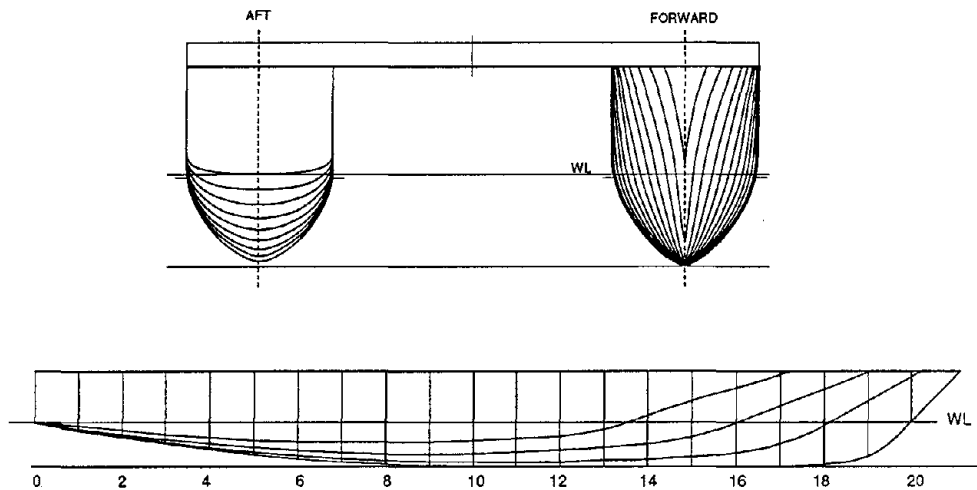


Figure 1.1: *Lines plane Catamaran 372.*

1.2.1 The still water resistance tests

A series of experiments have been carried out to obtain the resistance, trim and sinkage of the model over the forward speed range $Fn = 0.18$ to $Fn = 0.75$. An illustration of the experimental set-up is given in Figure 1.2.

The sinkage of the model was measured (using a potentiometer and a wire connected with the model) at the centre of gravity and in a second point which was positioned 679 mm behind the centre of gravity. From the two vertical displacements the trim angle can be calculated, while the sinkage at the centre of gravity was directly obtained from the measurements.

To obtain the stationary wave pattern created by the vessel at a certain forward speed, wave cut measurements were carried out. For this purpose a wave height meter was positioned in the towing tank at a fixed distance from the tank wall, which corresponded with a distance of 275 mm from the demihull centre line, see Figure 1.3. By passing the wave height meter the wave system generated by the vessel was recorded. Due to the reflections of the tank wall only the first part of the recorded wave is a valid representation of the steady wave system generated by the catamaran. An extra signal was recorded during the test run which was triggered by the carriage passing the wave height meter. Using this trigger pulse it was possible to obtain the relation between the vessel position and the recorded wave signal.

A grid consisting of the still waterline, two waterlines at a distance of 50 and 100 mm below the still waterline, two waterlines at a distance of 50 and 100 mm above the

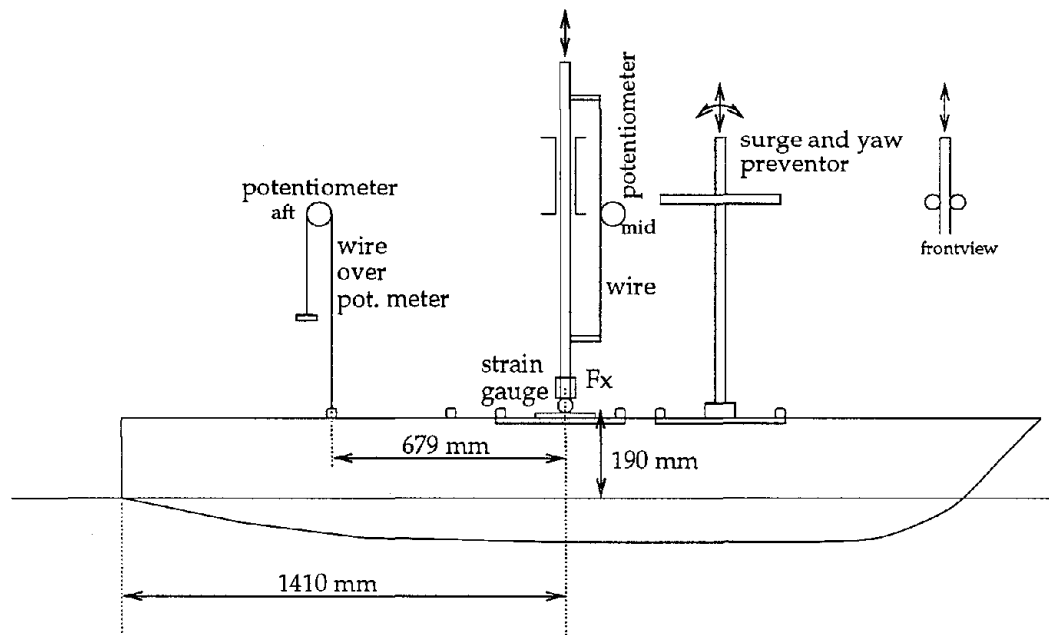


Figure 1.2: *Experimental set-up for wave resistance, trim and sinkage measurements and for the heave and pitch motion tests in head waves*

still waterline, and vertical lines at equidistant distance of 100 mm starting at the fore perpendicular were drawn on both demihulls at port side. During the test runs a series of pictures was taken which were afterwards used to construct the steady wave contour on the hull.

1.2.2 The heave and pitch motion tests

To obtain the heave and pitch response amplitude operators in head waves a series of motion tests have been carried out. An illustration of the experimental set-up for these tests is presented in Figure 1.2, which is the same set-up as for the resistance tests.

The vertical displacement in the centre of gravity and in a point 679 mm behind the centre of gravity were recorded using potentiometers. From the two displacements the heave and pitch response could be calculated. The phase lag between the responses and the incoming wave system was obtained using as reference signal the signal of a wave height meter which was positioned in line with the centre of gravity.

The model was tested at four different forward speed values, that is $F_n = 0.30$, $F_n = 0.45$, $F_n = 0.60$ and $F_n = 0.75$, and at each forward speed in several wave frequency

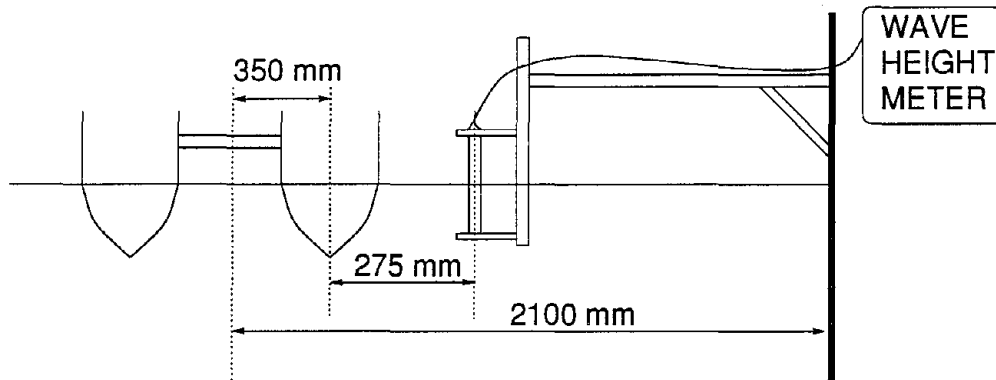


Figure 1.3: *Wave cut measuring layout*

conditions. A second wave height meter, positioned in front of the model, was used to obtain the undisturbed incoming wave signal.

1.2.3 The forced heave and pitch oscillation tests

To obtain the heave and pitch added mass and fluid damping values, forced heave and pitch oscillation tests have been carried out. An illustration of the experimental set-up is presented in Figure 1.4.

The oscillator and the model were connected using two load cells at each connection point, measuring the force in the vertical and horizontal direction. The oscillations were carried out at four different forward speed values, that is $F_n = 0.30$, $F_n = 0.45$, $F_n = 0.60$ and $F_n = 0.75$, and at each velocity for several oscillation frequencies ranging from $\omega = 3.20$ rad/sec up to $\omega = 5.85$ rad/sec. During these tests the trim and sinkage of the model were zero (the still water reference position of the model).

To investigate the influence of the trim and sinkage in the hydrodynamic terms an extra series of forced heave and pitch oscillation tests was performed in which the reference position of the vessel was changed using the trim and sinkage obtained from the still water resistance test. These experiments were only performed at $F_n = 0.75$.

During the forced heave and pitch oscillation tests the weight of the model was kept as light as possible, which had its influence on the pitch radius of gyration. In the calculation procedure the measured forces are corrected with the actual weight and gyration radius, see also Table 1.1.

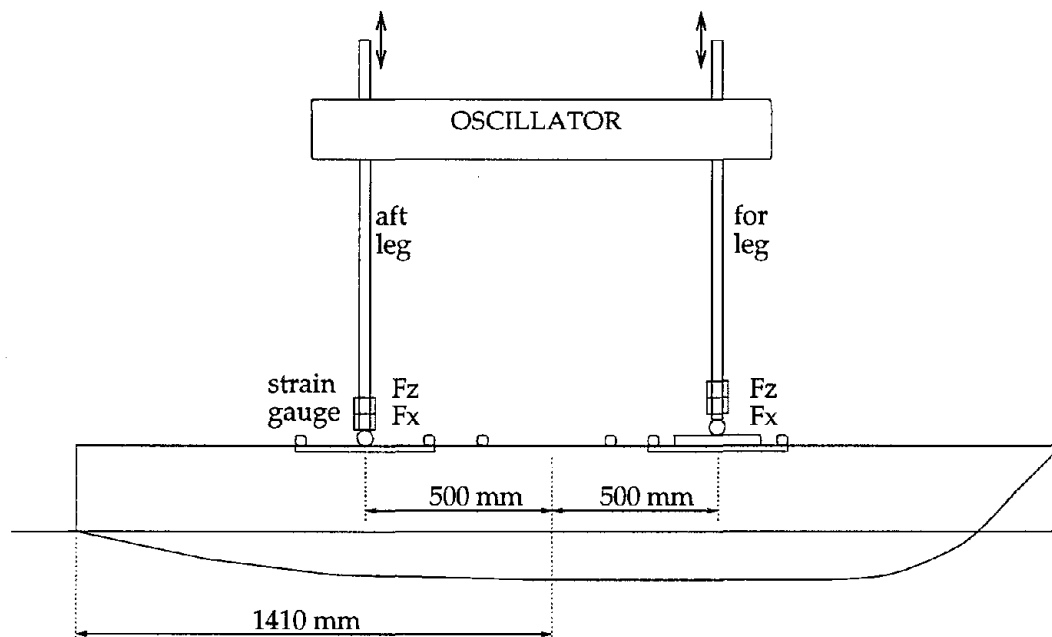


Figure 1.4: *Experimental set-up for forced heave and pitch oscillation tests and for the wave load tests*

1.2.4 The wave load tests

To obtain the wave loads on the catamaran vessel, restrained model tests have been carried out in head waves. The vertical and the horizontal forces at the two connecting points were measured. An illustration of the experimental set-up is given in Figure 1.4, which is the same set-up as for the oscillation tests (only now the oscillator is fixed).

The model tests were performed at four different forward velocities, $F_n = 0.30, 0.45, 0.60$ and 0.75 , and at each velocity a set wave frequencies was used, ranging from $\lambda/L = 0.6$ up to $\lambda/L = 2.0$. During the experiments the trim and sinkage of the model were zero. A wave height meter in the centre of gravity plane of the vessel was used to obtain the phase lag between the forces and the incoming wave system. A second wave height meter in front of the model to register the undisturbed incoming wave system.

The model weight of the model during the restrained model tests was kept equal to the model weight in the forced oscillation tests. This is possible since the model was forced in it's reference position (zero trim and sinkage) by the oscillator.

1.3 Test program overview

In Table 1.2 through 1.7 an overview is given of all the runs presented in this report.

run	U [m/s]	Fn	run	U [m/s]	Fn
2	1.630	0.300	14	2.709	0.499
4	2.440	0.450	16	2.978	0.549
6	3.258	0.601	18	3.527	0.650
8	4.058	0.748	20	3.797	0.700
10	1.890	0.348	22	0.999	0.184
12	2.170	0.400	24	1.300	0.240

Table 1.2: *Test run overview of wave resistance measurement program*

run	Fn	λ/L	run	Fn	λ/L	run	Fn	λ/L	run	Fn	λ/L
26	0.30	0.601	50	0.45	0.601	72	0.60	0.798	98	0.75	0.998
28	0.30	0.803	52	0.45	0.799	74	0.60	0.897	100	0.75	1.100
30	0.30	0.899	54	0.45	0.899	76	0.60	0.999	102	0.75	1.204
32	0.30	0.965	56	0.45	1.001	78	0.60	1.098	104	0.75	1.297
34	0.30	1.007	58	0.45	1.097	80	0.60	1.200	106	0.75	1.394
36	0.30	1.101	60	0.45	1.201	82	0.60	1.292	108	0.75	1.491
38	0.30	1.204	62	0.45	1.295	84	0.60	1.393	110	0.75	1.596
40	0.30	1.301	64	0.45	1.396	86	0.60	1.490	112	0.75	1.806
42	0.30	1.395	66	0.45	1.595	88	0.60	1.600	114	0.75	1.985
44	0.30	1.596	68	0.45	1.803	90	0.60	1.795	116	0.75	1.694
46	0.30	1.803	70	0.45	1.991	96	0.60	1.977	118	0.75	1.906
48	0.30	1.978				122	0.60	1.741	120	0.75	2.012
126	0.30	0.699				124	0.60	1.816	128	0.75	1.983

Table 1.3: *Test run overview of the heave and pitch motion measurement program*

run	F_n	ω	run	F_n	ω	run	F_n	ω	run	F_n	ω
301	0.30	3.267	319	0.45	3.230	339	0.60	3.194	355	0.75	3.196
303	0.30	3.577	321	0.45	3.584	341	0.60	3.568	357	0.75	3.351
305	0.30	3.838	323	0.45	3.847	343	0.60	3.686	359	0.75	3.557
307	0.30	3.958	325	0.45	3.988	345	0.60	3.783	361	0.75	3.664
309	0.30	4.123	327	0.45	4.106	347	0.60	3.954	363	0.75	3.816
311	0.30	4.309	329	0.45	4.288	349	0.60	4.131	365	0.75	3.960
313	0.30	4.517	331	0.45	4.512	351	0.60	4.489	367	0.75	4.103
315	0.30	4.762	333	0.45	4.756	353	0.60	5.060	369	0.75	4.519
317	0.30	5.046	335	0.45	5.044				371	0.75	5.064
			337	0.45	5.827						
			455	0.45	5.846						
			457	0.45	5.383						

Table 1.4: *Test run overview of forced heave oscillation measurement program*

run	F_n	ω	run	F_n	ω	run	F_n	ω	run	F_n	ω
373	0.30	3.261	393	0.45	3.220	413	0.60	3.205	429	0.75	3.191
375	0.30	3.627	395	0.45	3.572	415	0.60	3.378	431	0.75	3.566
377	0.30	3.839	397	0.45	3.808	417	0.60	3.577	433	0.75	3.371
379	0.30	3.977	399	0.45	3.958	419	0.60	3.752	435	0.75	3.677
381	0.30	4.116	401	0.45	4.111	421	0.60	3.960	437	0.75	3.808
383	0.30	4.304	403	0.45	4.300	423	0.60	4.111	439	0.75	3.955
385	0.30	4.516	405	0.45	4.508	425	0.60	4.512	441	0.75	4.111
387	0.30	4.768	407	0.45	4.757	427	0.60	5.040	443	0.75	4.515
389	0.30	5.040	409	0.45	5.051				445	0.75	5.071
391	0.30	5.383	411	0.45	5.410						
			447	0.45	3.387						

Table 1.5: *Test run overview of forced pitch oscillation measurement program*

run	F_n	ω	run	F_n	ω
582	0.75	3.207	598	0.75	3.213
584	0.75	3.823	600	0.75	3.373
586	0.75	4.508	602	0.75	3.564
588	0.75	4.114	604	0.75	3.816
590	0.75	3.569	606	0.75	4.111
592	0.75	3.373			
594	0.75	5.071			

Table 1.6: *Test run overview of forced heave and pitch oscillation measurement program, with trim and sinkage correction*

run	F_n	λ	run	F_n	λ	run	F_n	λ	run	F_n	λ
502	0.30	1.812	504	0.45	1.806	540	0.60	2.400	558	0.75	2.400
510	0.30	2.412	508	0.45	2.394	542	0.60	2.724	560	0.75	2.994
512	0.30	3.000	514	0.45	3.003	544	0.60	3.006	562	0.75	2.697
516	0.30	3.294	518	0.45	3.327	546	0.60	3.591	564	0.75	3.606
520	0.30	3.597	522	0.45	3.585	548	0.60	3.891	566	0.75	3.885
524	0.30	3.897	526	0.45	3.888	550	0.60	4.188	568	0.75	4.203
528	0.30	4.197	530	0.45	4.197	552	0.60	4.800	570	0.75	4.797
532	0.30	4.791	534	0.45	4.779	554	0.60	5.307	572	0.75	5.958
536	0.30	5.967	538	0.45	5.919	556	0.60	5.940	578	0.75	3.291
574	0.30	2.055	576	0.45	2.055						

Table 1.7: *Test run overview of wave load measurement program*

x=-15.00		x=-14.25		x=-13.50		x=-12.75		x=-12.00		x=-11.25	
z	y	z	y	z	y	z	y	z	y	z	y
2.00	1.20	2.00	1.20	2.00	1.20	2.00	1.20	2.00	1.20	2.00	1.20
1.90	1.20	1.83	1.20	1.80	1.198	1.80	1.20	1.80	1.199	1.78	1.20
1.80	1.193	1.661	1.15	1.60	1.141	1.578	1.15	1.60	1.170	1.509	1.15
1.70	1.141	1.551	1.05	1.50	1.065	1.425	1.05	1.50	1.132	1.321	1.05
1.57	1.00	1.469	0.90	1.40	0.936	1.301	0.90	1.40	1.072	1.160	0.90
1.545	0.80	1.427	0.75	1.339	0.80	1.228	0.75	1.227	0.90	1.059	0.75
1.492	0.60	1.405	0.60	1.293	0.60	1.187	0.60	1.088	0.60	0.995	0.60
1.475	0.30	1.381	0.30	1.260	0.30	1.144	0.30	1.033	0.30	0.928	0.30
1.470	0.00	1.372	0.00	1.247	0.00	1.127	0.00	1.011	0.00	0.901	0.00

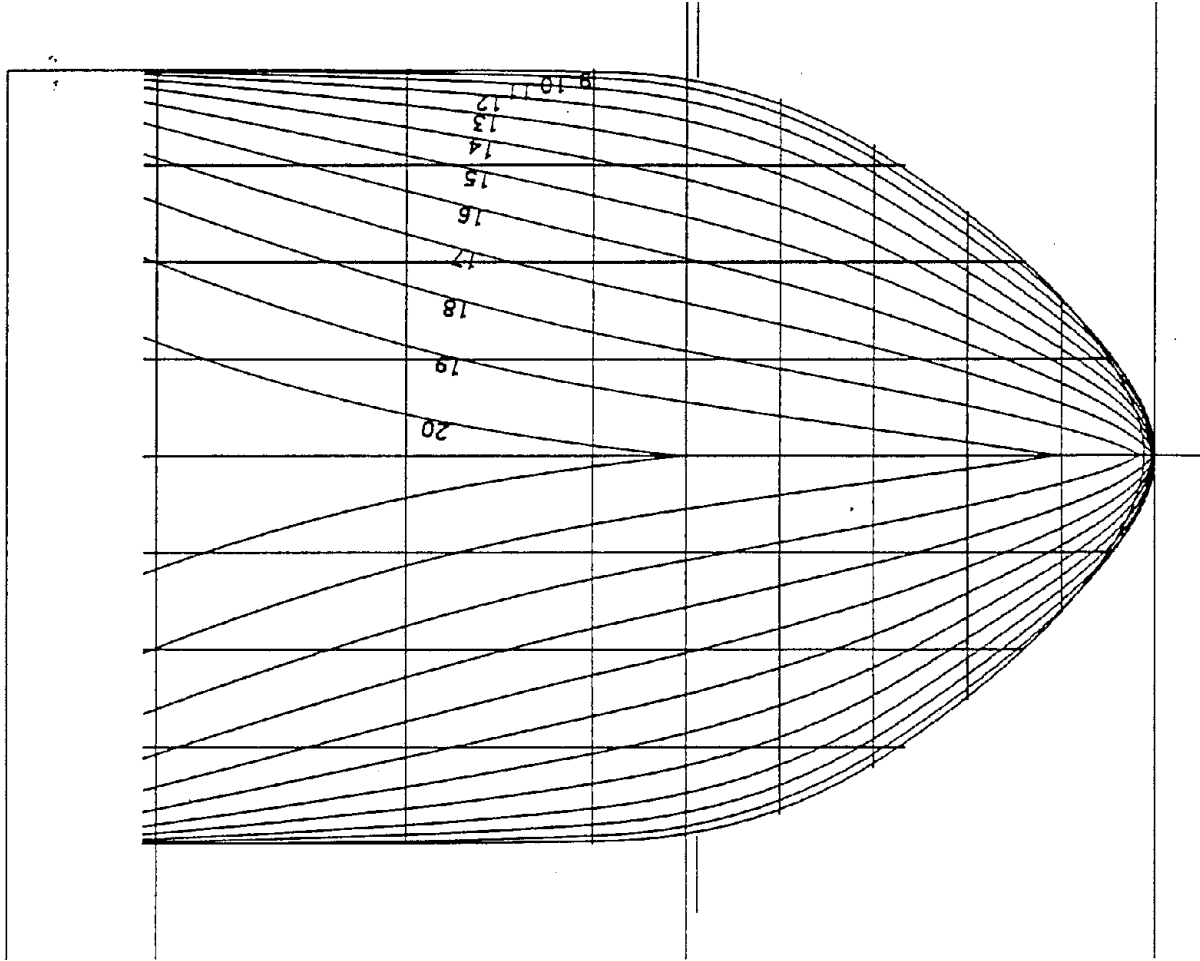
x=-10.50		x=-9.00		x=-7.50		x=-6.00		x=-4.50		x=-3.00	
z	y	z	y	z	y	z	y	z	y	z	y
2.00	1.20	2.00	1.20	2.00	1.20	2.00	1.20	2.00	1.20	2.00	1.20
1.80	1.199	1.80	1.20	1.80	1.20	1.70	1.20	1.70	1.20	1.70	1.20
1.60	1.182	1.70	1.198	1.60	1.194	1.50	1.186	1.50	1.187	1.50	1.187
1.50	1.157	1.50	1.170	1.50	1.180	1.30	1.138	1.30	1.143	1.30	1.138
1.30	1.066	1.30	1.104	1.40	1.157	1.10	1.057	1.10	1.067	1.10	1.067
1.098	0.90	1.10	0.987	1.20	1.084	0.90	0.938	0.90	0.960	0.90	0.963
0.909	0.60	0.90	0.808	1.00	0.969	0.70	0.780	0.70	0.817	0.70	0.830
0.827	0.30	0.757	0.60	0.80	0.805	0.50	0.553	0.50	0.638	0.50	0.670
0.795	0.00	0.645	0.30	0.632	0.60	0.361	0.30	0.40	0.521	0.30	0.459
		0.60	0.00	0.489	0.30	0.30	0.124	0.260	0.30	0.189	0.30
				0.429	0.00	0.284	0.00	0.20	0.163	0.136	0.20
								0.169	0.00	0.099	0.10
										0.085	0.00

x=-1.50		x=0.00		x=1.50		x=3.00		x=4.50		x=6.00	
z	y	z	y	z	y	z	y	z	y	z	y
2.00	1.194	2.00	1.179	2.00	1.151	2.00	1.106	2.00	1.040	2.00	0.953
1.80	1.191	1.80	1.172	1.80	1.138	1.80	1.086	1.80	1.012	1.80	0.917
1.50	1.168	1.50	1.142	1.50	1.100	1.50	1.039	1.50	0.957	1.50	0.855
1.20	1.091	1.20	1.061	1.20	1.016	1.20	0.953	1.20	0.870	1.20	0.771
0.90	0.949	0.90	0.919	0.90	0.874	0.90	0.816	0.90	0.742	0.90	0.655
0.60	0.747	0.60	0.720	0.60	0.679	0.60	0.628	0.60	0.569	0.60	0.502
0.30	0.485	0.30	0.479	0.30	0.453	0.30	0.416	0.30	0.375	0.30	0.329
0.148	0.30	0.133	0.30	0.139	0.30	0.161	0.30	0.090	0.20	0.119	0.20
0.089	0.20	0.066	0.20	0.063	0.20	0.072	0.20	0.025	0.10	0.035	0.10
0.048	0.10	0.023	0.10	0.015	0.10	0.018	0.10	0.00	0.00	0.00	0.00
0.034	0.00	0.009	0.00	0.00	0.00	0.00	0.00				

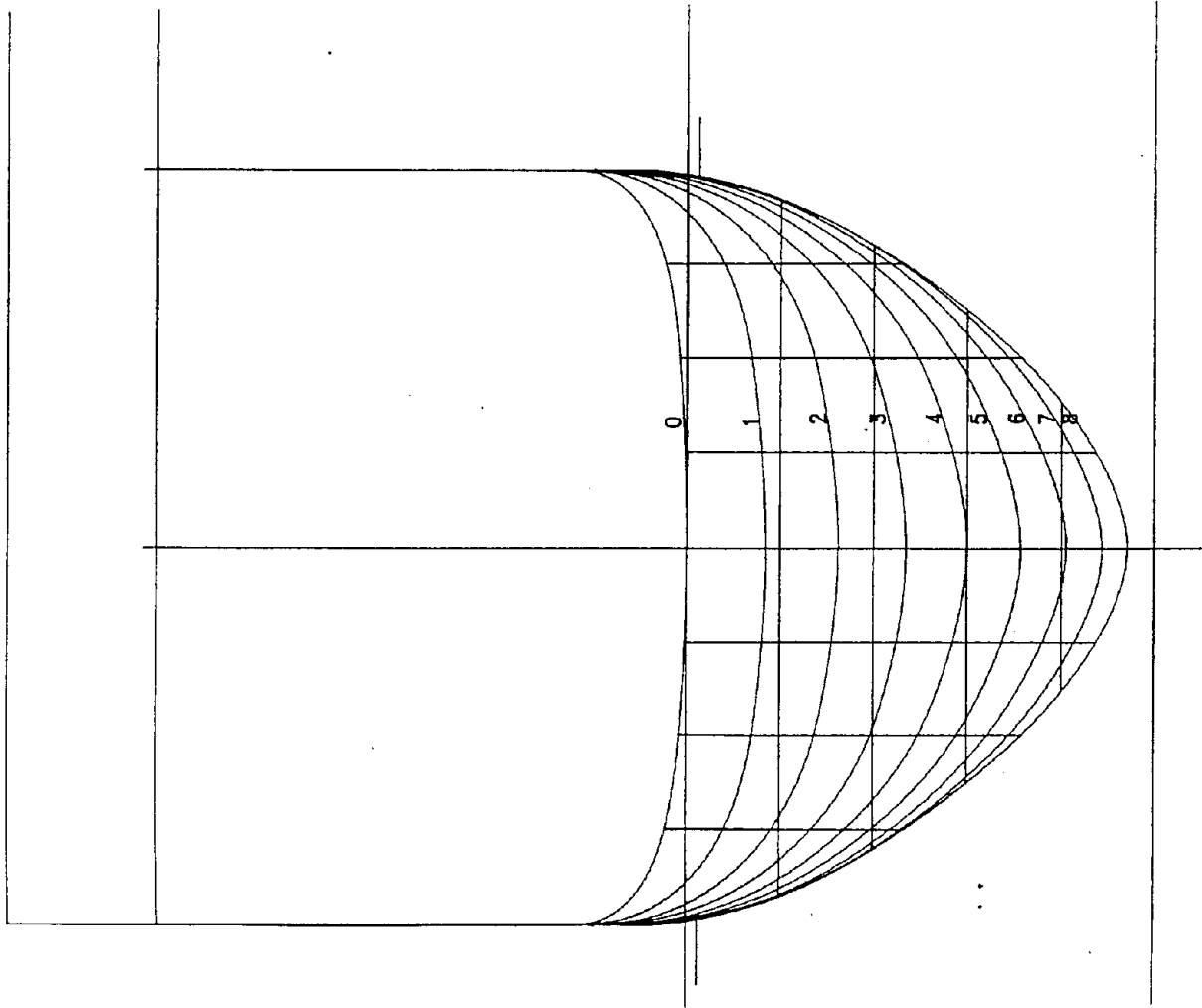
x=7.50		x=9.00		x=10.50		x=12.00		x=13.50		x=bow	
z	y	z	y	z	y	z	y	z	y	x	z
2.00	0.848	2.00	0.724	2.00	0.584	2.00	0.426	2.00	0.250	15.435	2.00
1.80	0.805	1.80	0.678	1.80	0.537	1.80	0.382	1.80	0.213	15.00	1.525
1.50	0.739	1.50	0.610	1.50	0.472	1.50	0.324	1.50	0.166	14.50	1.018
1.20	0.658	1.20	0.536	1.20	0.407	1.20	0.269	1.20	0.123	14.00	0.600
0.90	0.557	0.90	0.450	0.90	0.336	1.00	0.232	0.90	0.082		
0.60	0.428	0.60	0.347	0.60	0.256	0.80	0.195	0.60	0.041		
0.30	0.280	0.30	0.224	0.30	0.161	0.60	0.155	0.317	0.00		
0.10	0.151	0.10	0.116	0.146	0.10	0.40	0.110				
0.00	0.00	0.00	0.00	0.002	0.00	0.20	0.059				
						0.049	0.00				

Table 1.8: List of offsets

FORWARD



AFT



2 Measurements results

2.1 The still water resistance tests

Table:

2.1: Still water resistance, sinkage and trim, $F_n = 0.18$ to 0.75

Figure:

2.1: Still water resistance, $F_n = 0.18$ to 0.75

2.2: Trim angle and sinkage, $F_n = 0.18$ to 0.75

2.3: Longitudinal wave cuts at $y/b = 1.146$, $F_n = 0.30, 0.45, 0.60$ and 0.75

2.4: Longitudinal wave cuts at $y/b = 1.146$, $F_n = 0.35, 0.50, 0.55$ and 0.65

2.5: Hull wave profile at $F_n = 0.30$

2.6: Hull wave profile at $F_n = 0.45$

The sinkage is defined positive if the centre of gravity moves upwards. The trim is defined negative if the bow moves upwards in relation to the centre of gravity. This is according the right handed coordinate system where the x- axis is positive forward, the y-axis is positive to port and the z-axis is positive upwards.

F_n	U [m/sec]	R [N]	sinkage [mm]	trim [deg]
0.184	0.999	3.31	-1.77	-0.035
0.240	1.300	6.60	-3.09	-0.048
0.300	1.627	12.77	-5.88	-0.091
0.348	1.890	16.14	-8.06	-0.050
0.400	2.170	26.23	-12.57	-0.48
0.450	2.439	41.75	-15.58	-1.39
0.499	2.709	52.79	-15.16	-1.99
0.549	2.978	59.08	-12.36	-2.18
0.601	3.258	63.50	-7.89	-2.09
0.650	3.527	66.00	-4.35	-2.01
0.700	3.797	70.60	-2.39	-1.90
0.748	4.058	74.71	+2.57	-1.35

Table 2.1: Still water resistance, sinkage and trim; $F_n = 0.18$ to $F_n = 0.75$.

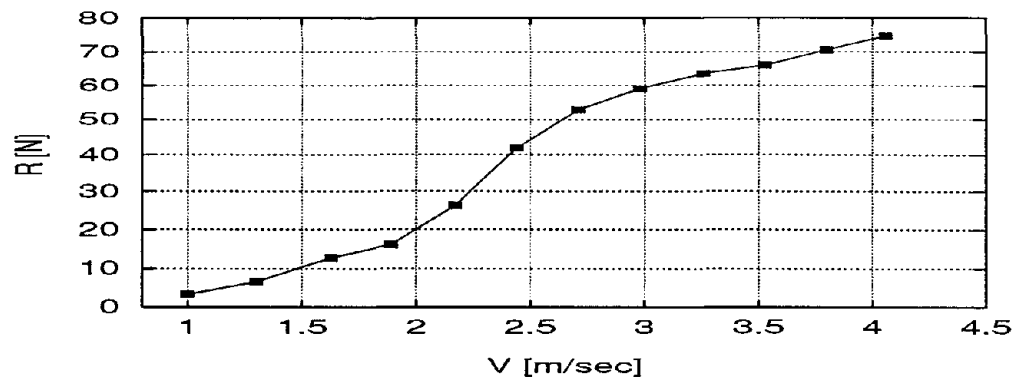


Figure 2.1: Still water resistance, $Fn = 0.18$ to 0.75 .

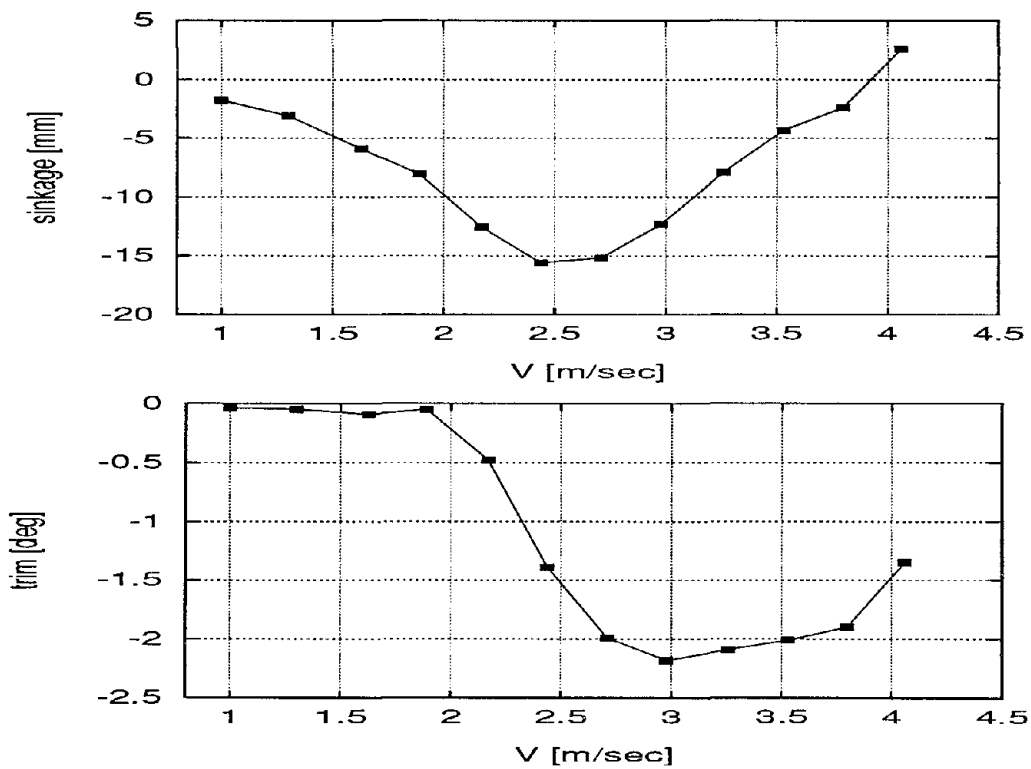


Figure 2.2: Sinkage at centre of gravity, and Trim angle, $Fn = 0.18$ to 0.75 .

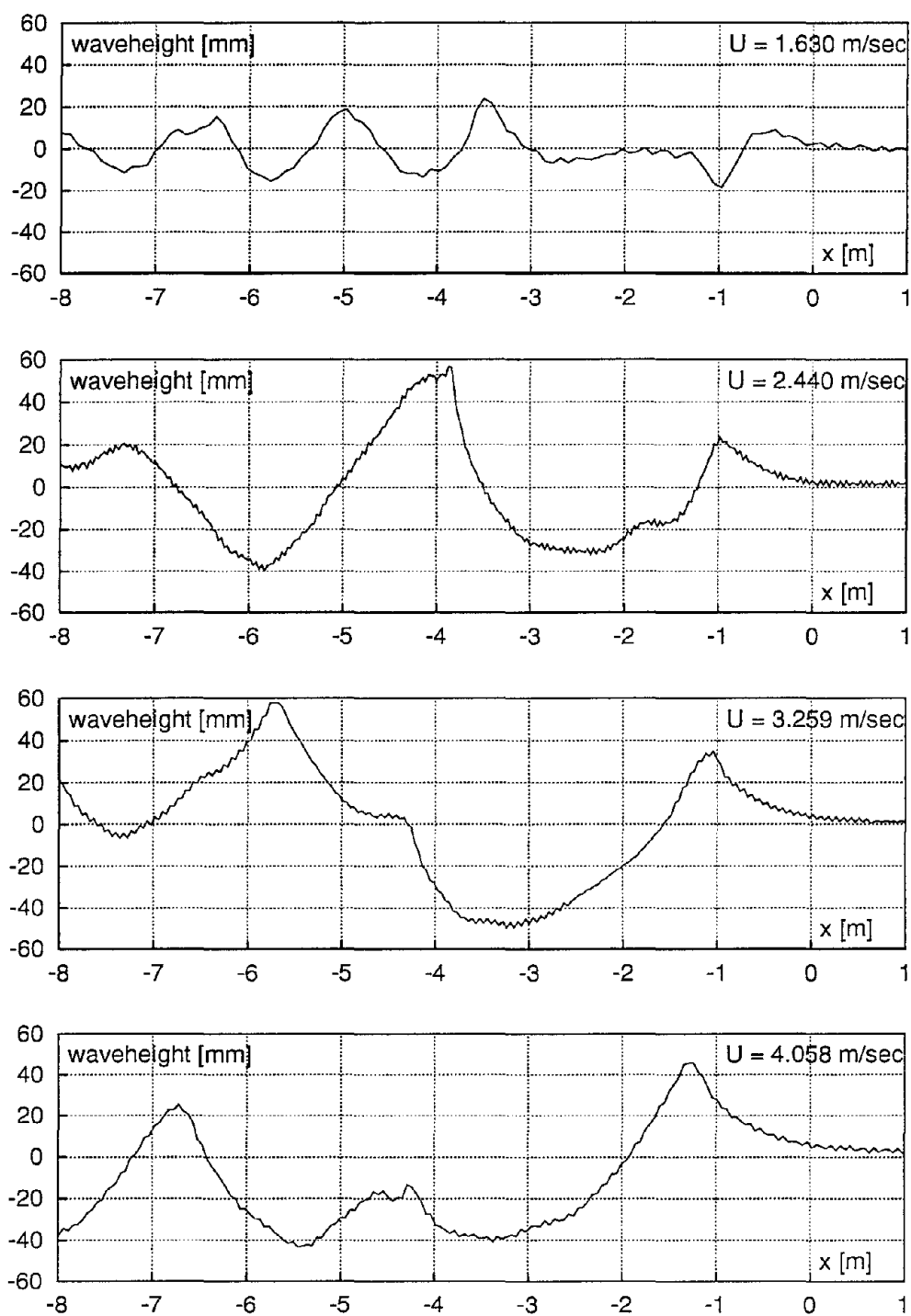


Figure 2.3: Longitudinal wave cuts at $y/b = 1.146$, $Fn = 0.30, 0.45, 0.60$ and 0.75 , Coordinate reference, $x = 0.0$ m = FP, $x = -3.0$ m = AP.

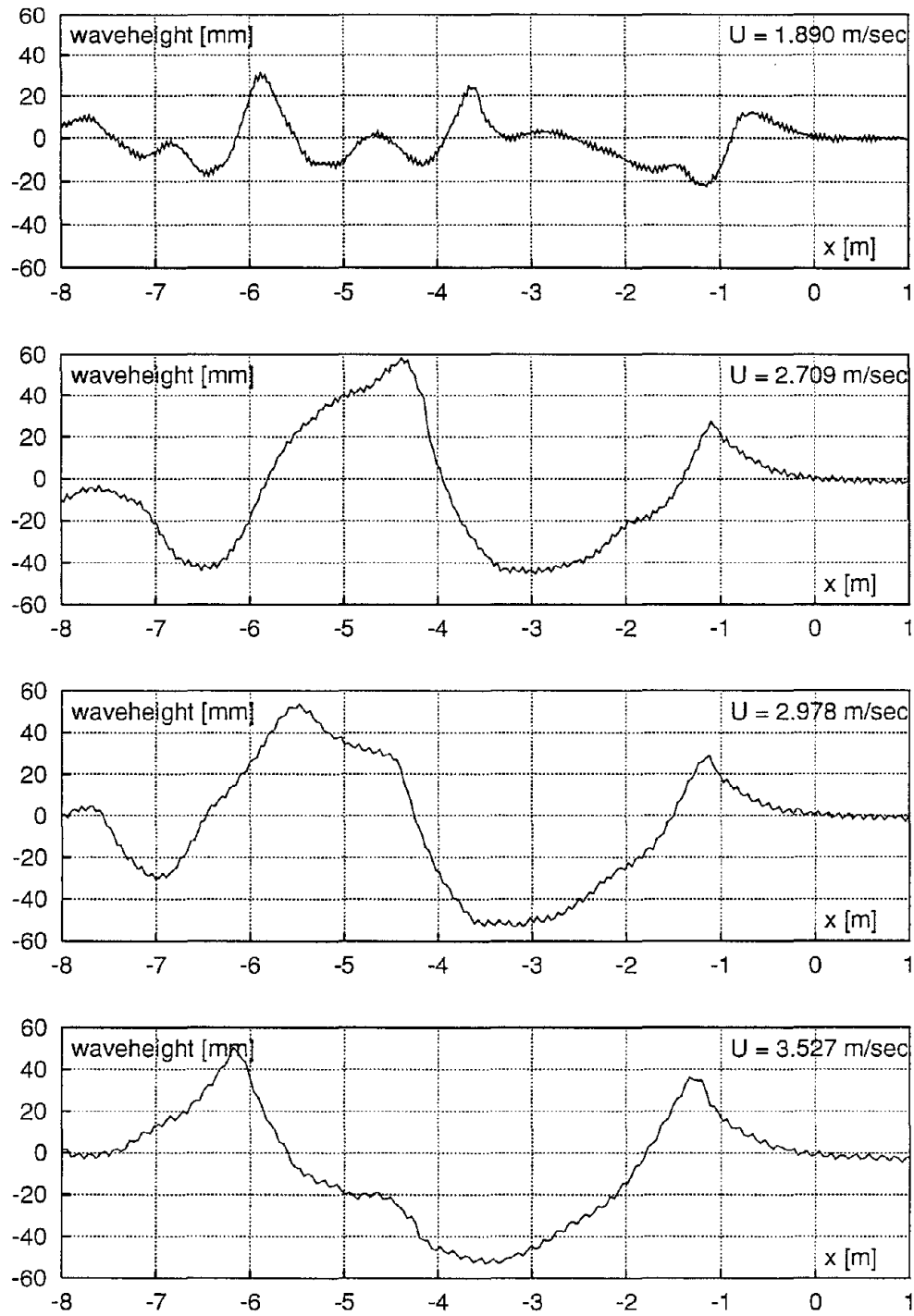


Figure 2.4: Longitudinal wave cuts at $y/b = 1.146$, $Fn = 0.35, 0.50, 0.55$ and 0.65 . Coordinate reference, $x = 0.0$ m = FP, $x = -3.0$ m = AP.

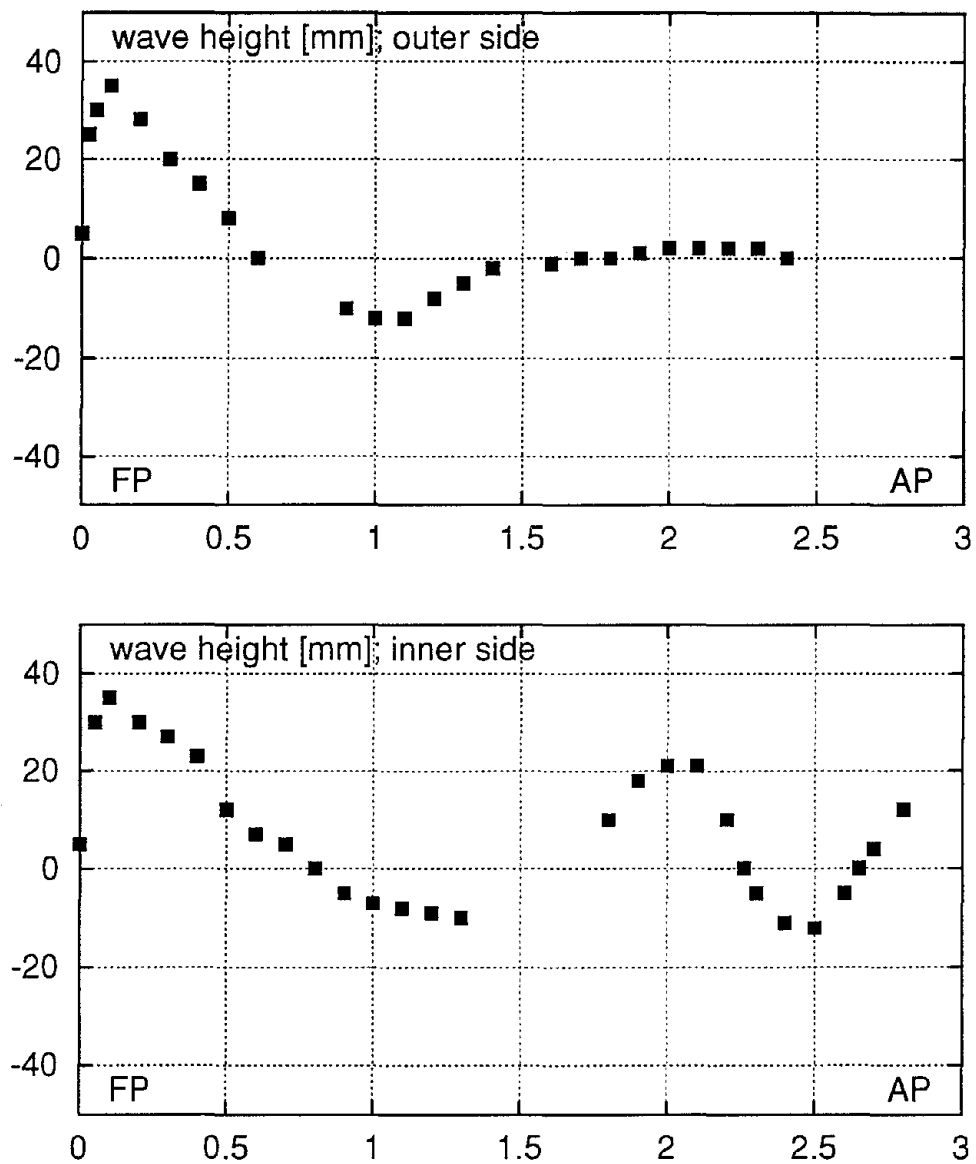


Figure 2.5: *Hull wave profile, obtained via photo's, $FN = 0.30$.*

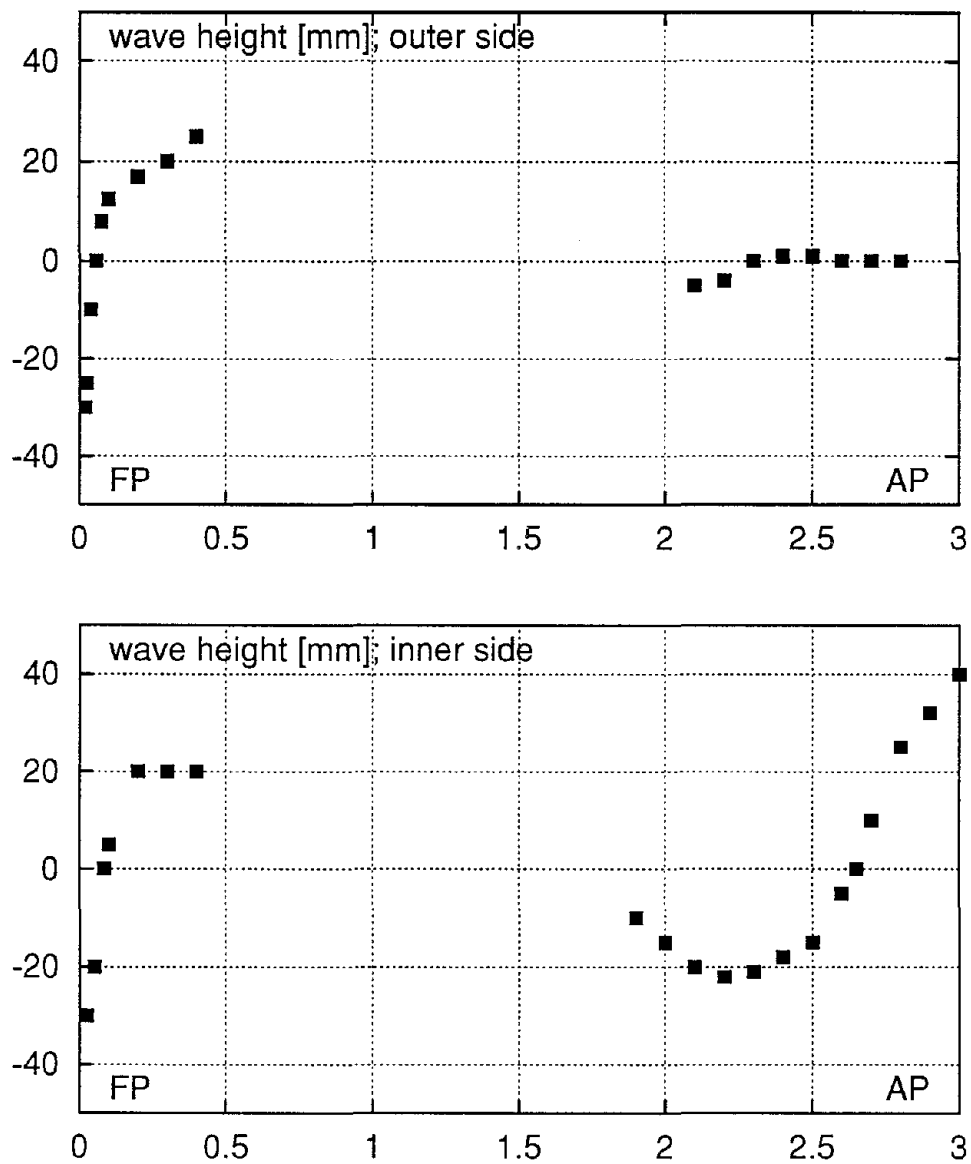


Figure 2.6: Hull wave profile, obtained via photo's, $FN = 0.45$.

2.2 The heave and pitch motion tests

Table:

2.2: Heave and Pitch response at $Fn = 0.30$

2.3: Heave and Pitch response at $Fn = 0.45$

2.4: Heave and Pitch response at $Fn = 0.60$

2.5: Heave and Pitch response at $Fn = 0.75$

Figure:

2.7: Heave and Pitch Response, $Fn = 0.30$

2.8: Heave and Pitch Response, $Fn = 0.45$

2.9: Heave and Pitch Response, $Fn = 0.60$

2.10: Heave and Pitch Response, $Fn = 0.75$

The heave (η_3, ϵ_3) and pitch (η_5, ϵ_5) motions are measured at the Froude numbers 0.30, 0.45, 0.60 and 0.75. During the experiments the motions of the model were registered in two points, according to Figure 1.2. The heave results are directly obtained for the measurements in the midpoint, while the pitch motions in the centre of gravity had to be calculated using $\bar{\eta}_5 = (\bar{z}_m - \bar{z}_a)/679$, where $\bar{z}_m$ and $\bar{z}_a$ are the complex motions in the midpoint (m) and the aft point (a). This complex amplitude provides the pitch amplitude ($\eta_5 = \text{mod } \bar{\eta}_5$) and the pitch phase ($\epsilon_5 = \arg \bar{\eta}_5$).

The motions are made dimensionless using the wave amplitude (ζ_a) and the wave number ($k = \omega^2/g$).

λ/L	ζ_a [mm]	η_3/ζ_a	ϵ_3 [deg]	$\eta_5/k\zeta_a$	ϵ_5 [deg]
0.601	21.973	0.144	28.835	0.055	51.631
0.699	19.712	0.076	-176.178	0.213	124.214
0.803	20.396	0.704	-89.729	0.464	157.208
0.899	19.686	1.268	-39.119	0.575	-179.152
0.965	19.224	1.358	-6.242	0.654	-169.111
1.007	19.267	1.327	-8.567	0.735	179.676
1.101	18.733	1.152	18.918	1.066	-161.644
1.204	18.598	0.823	20.385	1.229	-146.690
1.301	17.919	0.696	13.344	1.228	-135.056
1.395	18.035	0.680	4.620	1.189	-127.338
1.596	18.497	0.738	-0.936	1.129	-116.628
1.803	18.923	0.795	-2.212	1.094	-110.617
1.978	19.099	0.843	-1.160	1.075	-105.812

Table 2.2: Heave and Pitch response at $Fn = 0.30$.

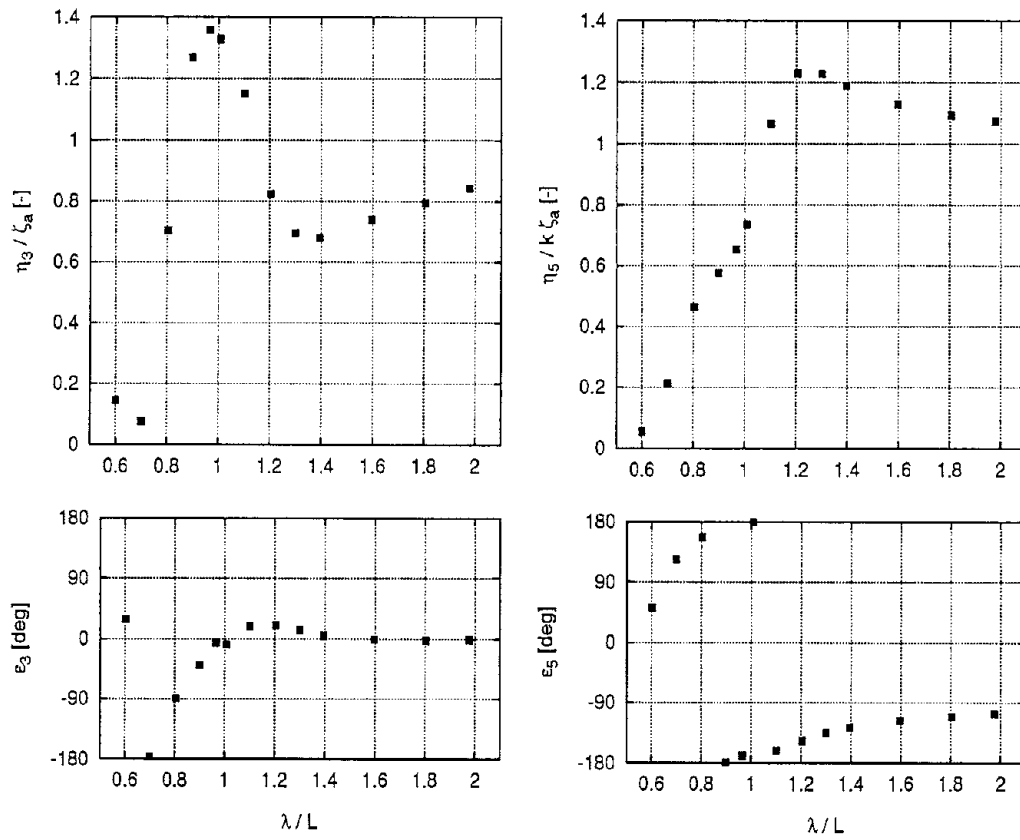


Figure 2.7: Heave and Pitch Response, $Fn = 0.30$.

λ/L	ζ_a [mm]	η_3/ζ_a	ϵ_3 [deg]	$\eta_5/k\zeta_a$	ϵ_5 [deg]
0.601	23.265	0.069	48.226	0.020	48.240
0.799	19.970	0.163	-156.039	0.182	120.531
0.899	18.836	0.522	-126.072	0.359	133.698
1.001	18.559	1.114	-96.491	0.590	152.149
1.097	17.899	1.769	-67.299	0.831	167.384
1.201	17.813	2.077	-39.698	1.023	178.879
1.295	17.456	1.999	-19.062	1.176	-169.778
1.396	18.039	1.675	-5.386	1.231	-159.423
1.595	17.409	1.300	6.051	1.354	-141.228
1.803	17.858	1.092	6.528	1.330	-128.509
1.991	18.236	1.027	3.162	1.298	-137.595

Table 2.3: Heave and Pitch response at $Fn = 0.45$.

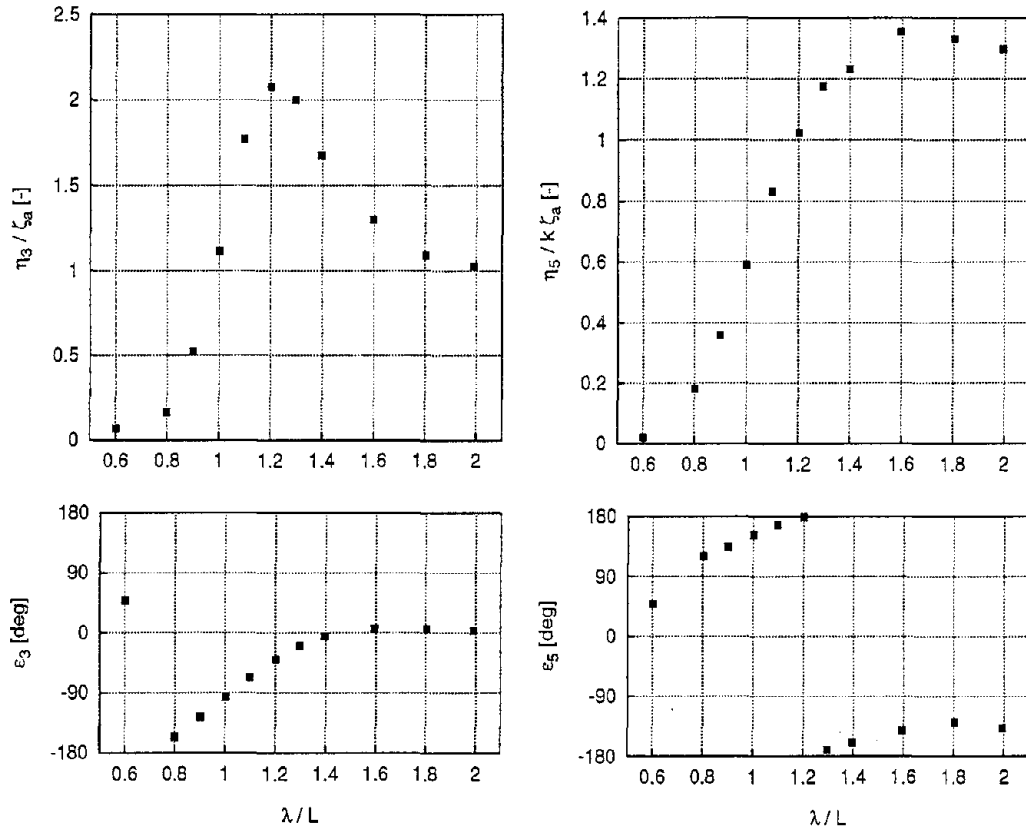


Figure 2.8: Heave and Pitch Response, $Fn = 0.45$.

λ/L	ζ_a [mm]	η_3/ζ_a	ϵ_3 [deg]	$\eta_5/k\zeta_a$	ϵ_5 [deg]
0.798	17.582	0.027	-137.879	0.089	102.710
0.897	19.223	0.145	-145.283	0.165	111.235
0.999	18.679	0.379	-137.034	0.304	121.156
1.098	18.297	0.749	-125.089	0.485	131.736
1.200	17.173	1.392	-104.830	0.735	149.084
1.292	17.560	1.892	-85.120	0.864	161.502
1.393	18.801	2.197	-66.036	0.932	168.045
1.490	19.151	2.370	-46.973	1.071	172.611
1.600	19.148	2.376	-29.403	1.299	-179.873
1.741	14.331	1.984	-9.500	1.535	-161.759
1.795	18.312	2.007	-4.306	1.657	-157.382
1.816	13.591	1.752	-0.797	1.587	-149.846
1.977	19.816	1.432	6.583	1.485	-137.595

Table 2.4: Heave and Pitch response at $Fn = 0.60$.

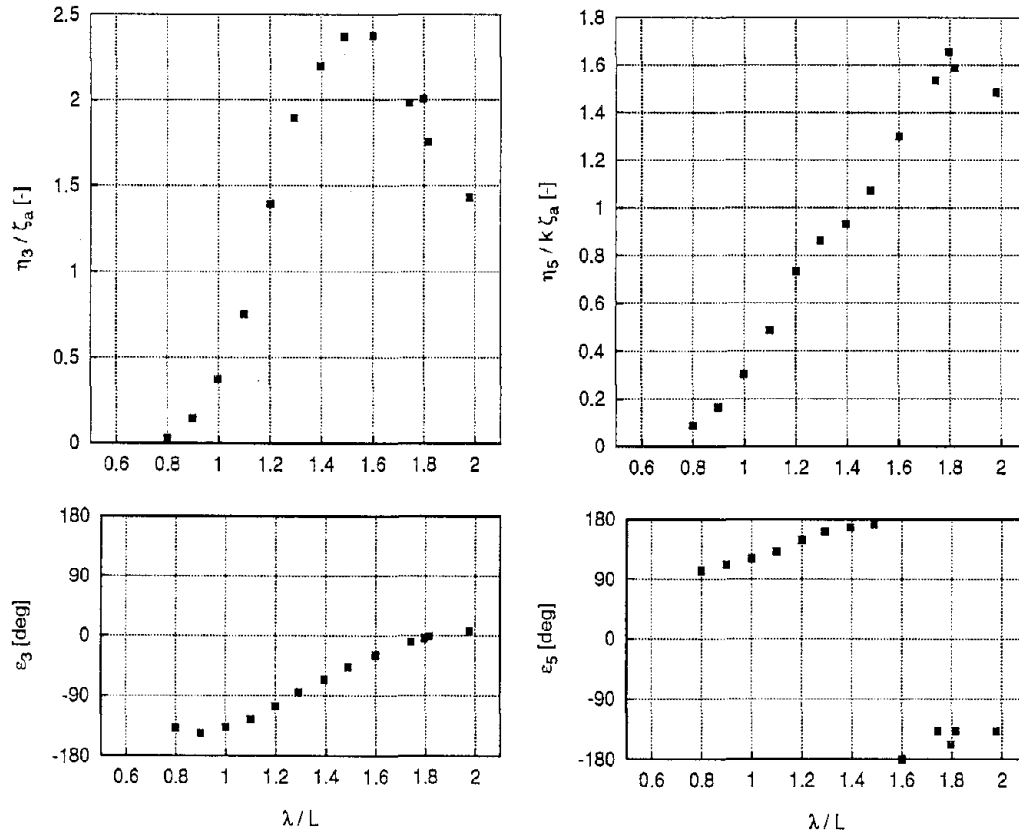


Figure 2.9: Heave and Pitch Response, $Fn = 0.60$.

λ/L	ζ_a [mm]	η_3/ζ_a	ϵ_3 [deg]	$\eta_5/k\zeta_a$	ϵ_5 [deg]
0.998	18.986	0.194	-145.897	0.172	113.207
1.100	17.959	0.378	-141.914	0.287	118.312
1.204	18.052	0.658	-130.823	0.437	127.927
1.297	17.839	1.016	-121.299	0.608	136.451
1.394	18.085	1.498	-109.074	0.801	145.608
1.491	17.960	2.026	-89.915	0.975	161.003
1.596	18.082	2.428	-73.224	1.078	171.383
1.694	14.001	2.495	-59.770	1.029	173.632
1.806	18.951	2.532	-47.209	1.088	174.751
1.906	14.189	2.493	-33.016	1.287	179.075
1.983	14.000	2.420	-19.895	1.482	-172.659
1.985	19.103	2.443	-19.916	1.437	-173.108
2.012	14.000	2.408	-20.848	1.503	-173.855

Table 2.5: Heave and Pitch response at $Fn = 0.75$.

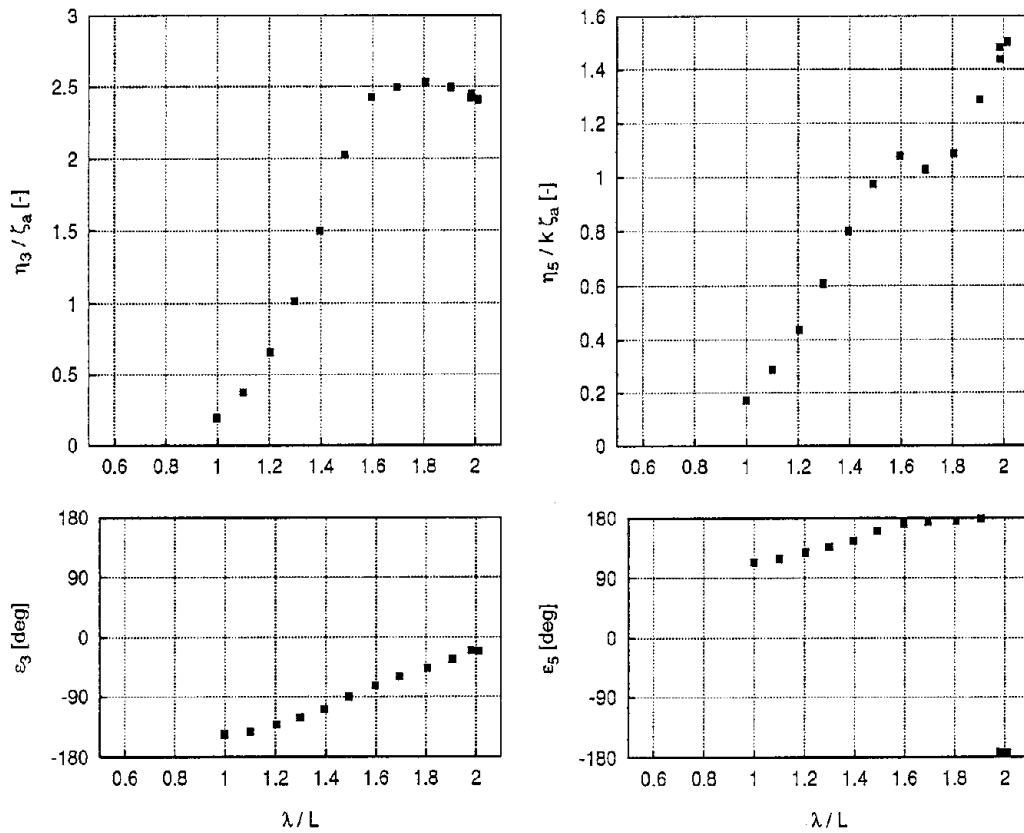


Figure 2.10: Heave and Pitch Response, $Fn = 0.75$.

2.3 The forced heave and pitch oscillation tests

Table:

- 2.7: Overview of static restoring coefficients experiments, zero trim and sinkage
- 2.8: Overview of static restoring coefficients experiments, trim and sinkage as with $F_n = 0.75$
- 2.9: Added mass and Fluid damping results, heave oscillation test, $F_n = 0.30$, zero trim and sinkage
- 2.10: Added mass and Fluid damping results, pitch oscillation test, $F_n = 0.30$, zero trim and sinkage
- 2.11: Added mass and Fluid damping results, heave oscillation test, $F_n = 0.45$, zero trim and sinkage
- 2.12: Added mass and Fluid damping results, pitch oscillation test, $F_n = 0.45$, zero trim and sinkage
- 2.13: Added mass and Fluid damping results, heave oscillation test, $F_n = 0.60$, zero trim and sinkage
- 2.14: Added mass and Fluid damping results, pitch oscillation test, $F_n = 0.60$, zero trim and sinkage
- 2.15: Added mass and Fluid damping results, heave oscillation test, $F_n = 0.75$, zero trim and sinkage
- 2.16: Added mass and Fluid damping results, pitch oscillation test, $F_n = 0.75$, zero trim and sinkage
- 2.17: Added mass and Fluid damping results, heave oscillation test, $F_n = 0.75$, trim and sinkage correction
- 2.18: Added mass and Fluid damping results, pitch oscillation test, $F_n = 0.75$, trim and sinkage correction

Figure:

- 2.11: Added mass values, $F_n = 0.30$
- 2.12: Fluid damping values, $F_n = 0.30$
- 2.13: Added mass values, $F_n = 0.45$
- 2.14: Fluid damping values, $F_n = 0.45$
- 2.15: Added mass values, $F_n = 0.60$
- 2.16: Fluid damping values, $F_n = 0.60$
- 2.17: Added mass values, $F_n = 0.75$
- 2.18: Fluid damping values, $F_n = 0.75$
- 2.19: Added mass values, $F_n = 0.75$, trim and sinkage correction
- 2.20: Fluid damping values, $F_n = 0.75$, trim and sinkage correction

2.3.1 Forced heave oscillation tests

In the forced heave oscillation tests the heave motion of the model is prescribed by

$$z = z_a \cos(\omega t) \quad (2.1)$$

where z_a is the prescribed heave motion amplitude, being 15 mm during all the experiments. For the heave oscillation test the equation of motion can be written as

$$\begin{aligned} (M + A_{33})\ddot{z} + B_{33}\dot{z} + C_{33}z &= F_{3a}\cos(\omega t + \epsilon_3) \\ A_{53}\ddot{z} + B_{53}\dot{z} + C_{53}z &= F_{5a}\cos(\omega t + \epsilon_5) \end{aligned} \quad (2.2)$$

where F_{3a} is the total force amplitude in the z direction and F_{5a} is the pitch moment amplitude around the centre of gravity. The heave to heave and heave to pitch coefficients can be deducted from the equation of motion, resulting in

$$\begin{aligned} A_{33} &= \frac{C_{33}}{\omega^2} - \frac{F_{3a}\cos\epsilon_3}{z_a\omega^2} - M \\ B_{33} &= \frac{F_{3a}\sin\epsilon_3}{z_a\omega} \\ A_{53} &= \frac{C_{53}}{\omega^2} - \frac{F_{5a}\cos\epsilon_5}{z_a\omega^2} \\ B_{53} &= \frac{F_{5a}\sin\epsilon_5}{z_a\omega} \end{aligned} \quad (2.3)$$

The force and moment amplitude and phase are calculated from the complex force (F_3) and moment (F_5) in the centre of gravity, which are calculated using the complex force in the fore (F_f) and aft (F_a) oscillation leg, that is:

$$\begin{aligned} F_3 &= F_a + F_f \\ F_5 &= F_a \cdot l + F_f \cdot l \end{aligned} \quad (2.4)$$

where l is the distance from the centre of gravity to the oscillation leg (thus $2l$ is the distance between the two force measurement points, thus l being 500 mm in the experiments).

2.3.2 Forced pitch oscillation tests

In the forced pitch oscillation tests the pitch motion is prescribed by

$$\theta = \theta_a \cos(\omega t) \quad (2.5)$$

In the experiments the amplitude of the fore and aft leg were kept the same, but with a phase lag of 180 degrees. This results in a pitch amplitude of 30 mm over 1000 mm, which is 1.718 degrees. The equation of motion can be written as

$$\begin{aligned} A_{35}\ddot{\theta} + B_{35}\dot{\theta} + C_{35}\theta &= F_{3a}\cos(\omega t + \epsilon_3) \\ (I_{55} + A_{55})\ddot{\theta} + B_{55}\dot{\theta} + C_{55}\theta &= F_{5a}\cos(\omega t + \epsilon_5) \end{aligned} \quad (2.6)$$

where F_{3a} is the total force amplitude in the z direction and F_{5a} is the pitch moment amplitude around the centre of gravity. The pitch to pitch and pitch to heave coefficients can be deducted from the equation of motion, resulting in

$$\begin{aligned} A_{35} &= \frac{C_{35}}{\omega^2} - \frac{F_{3a}\cos\epsilon_3}{\theta_a\omega^2} \\ B_{35} &= \frac{F_{3a}\sin\epsilon_3}{\theta_a\omega} \\ A_{55} &= \frac{C_{55}}{\omega^2} - \frac{F_{5a}\cos\epsilon_5}{\theta_a\omega^2} - I_{55} \\ B_{55} &= \frac{F_{5a}\sin\epsilon_5}{\theta_a\omega} \end{aligned} \quad (2.7)$$

The complex force F_3 and moment F_5 are again calculated from the forces measured in the fore and aft oscillation leg, that is

$$\begin{aligned} F_3 &= F_a + F_f \\ F_5 &= F_a \cdot l + F_f \cdot l \end{aligned} \quad (2.8)$$

2.3.3 Restoring coefficients

The restoring coefficients C_{33} , C_{35} , C_{53} and C_{55} play an important role in the calculation of the added mass coefficients, as can be seen in the equations above. At zero forward speed the coefficients can be calculated using the still waterline, that is:

$$\begin{aligned} C_{33} &= \rho g A_w \\ C_{35} &= \rho g S_w \\ C_{53} &= \rho g S_w \\ C_{55} &= \rho g \nabla G M_L \end{aligned} \quad (2.9)$$

where A_w is the water plane area, S_w is the first order moment of the water plane, and $G M_L$ is the longitudinal metacentric height. Using the ship geometry these values can

be calculated using:

$$\begin{aligned}
C_{33} &= \rho g \int_{x=0}^{x=L} y_w(x) dx \\
C_{35} &= \rho g \int_{x=0}^{x=L} x \cdot y_w(x) dx \\
C_{53} &= C_{35} \\
C_{55} &= \rho g \int_{x=0}^{x=L} x^2 \cdot y_w(x) dx
\end{aligned} \tag{2.10}$$

For the catamaran model the results of the calculations are presented in Table 2.6.

coefficient	value
C_{33}	10782 N/m
C_{35}	1850 N/m ²
C_{53}	1850 N/m ²
C_{55}	6144 N/m ³

Table 2.6: *Calculated restoring coefficients using the ship geometry ($F_n = 0.0$; zero trim and sinkage).*

The restoring coefficients can be calculated using the geometry of the vessel as shown above, but it is also possible to obtain the restoring coefficients from static experiments. If the vessel is given a fixed heave or pitch displacement the restoring coefficients follow from:

$$C_{33} = \frac{F_3}{z} \quad C_{53} = \frac{F_5}{z} \quad C_{35} = \frac{F_3}{\theta} \quad C_{55} = \frac{F_5}{\theta} \tag{2.11}$$

where F_3 and F_5 are respectively the heave and pitch forces measured, z is the given heave displacement and θ is the given pitch displacement.

If the static experiments are performed by $F_n = 0$ the zero forward speed restoring terms are found, which must be comparable to the restoring terms obtained from the geometrical quantities of the vessel.

Due to the forward speed the trim and sinkage of the vessel will change and a certain wave contour exist over the ship length. All these factors have there influence on the restoring coefficients. Therefor a small series of static experiments is performed in which the influence of the trim, sinkage and forward speed is looked at.

U [m/sec]	z_a [mm]	z_f [mm]	C_{33}	C_{53}	C_{35}	C_{55}
0.0	15.0	15.0	7316	57		
0.0	5.0	5.0	9871	1677		
0.0	-5.0	-5.0	11338	1831		
0.0	-15.0	-15.0	6890	149		
0.0	-15.0	15.0			2094	5918
0.0	-15.0	15.0			2095	5917
0.0	-10.0	10.0			1973	5351
0.0	-5.0	5.0			1872	5857
0.0	-5.0	5.0			1907	5885
0.0	5.0	-5.0			1488	5552
0.0	5.0	-5.0			1554	5582
0.0	10.0	-10.0			1070	5251
0.0	15.0	-15.0			799	5027
0.0	15.0	-15.0			886	5050
0.0	-5.0	5.0			3037	5900
4.06	-5.0	5.0			2791	4384

Table 2.7: Overview of static restoring coefficients experiments, zero trim and sinkage.

U [m/sec]	z_a [mm]	z_f [mm]	C_{33}	C_{53}	C_{35}	C_{55}
0.0	15.0	15.0	10356	2424		
0.0	10.0	10.0	10667	2480		
0.0	5.0	5.0	10545	2449		
0.0	-5.0	-5.0	10208	2326		
0.0	-10.0	-10.0	10422	2324		
0.0	-15.0	-15.0	10571	2305		
4.06	-5.0	-5.0	11892	867		
4.06	5.0	5.0	11494	896		
4.06	5.0	-5.0			4660	5466
4.06	-5.0	5.0			4398	5843

Table 2.8: Overview of static restoring coefficients experiments, all runs are with trim and sinkage as with $Fn = 0.75$.

In the calculations of the added mass values the restoring coefficients of Table 2.6 are used, that is by zero forward speed, zero sinkage and zero trim angle. The results are also plotted in Figure 2.11 and 2.12.

$\omega\sqrt{L/g}$	A_{33} [kg]	A_{53} [kg m]	B_{33} [kg/s]	B_{53} [kg m/s]
2.787	2.106	9.635	349.749	179.134
3.154	9.716	6.532	270.255	106.913
3.475	17.680	6.954	222.346	54.427
3.629	22.002	8.036	204.643	30.636
3.841	28.941	10.712	186.486	1.142
4.088	34.889	14.332	179.663	-24.319
4.371	42.894	19.746	190.841	-39.684
4.718	50.533	25.880	240.699	-23.362
5.130	53.902	29.212	341.584	40.571

Table 2.9: Added mass and Fluid damping results from heave oscillation test, $Fn = 0.30$, zero trim and sinkage.

$\omega\sqrt{L/g}$	A_{55} [kg m ²]	A_{35} [kg m]	B_{55} [kg m ² /s]	B_{35} [kg m/s]
2.780	64.994	24.161	113.474	120.30
3.213	61.197	24.618	137.288	169.97
3.475	59.219	23.509	156.149	198.76
3.652	57.540	22.295	171.590	218.05
3.833	55.737	20.380	192.745	238.07
4.082	52.433	17.221	222.011	261.62
4.368	47.232	12.760	248.854	275.71
4.725	40.854	7.853	250.246	258.86
5.119	37.295	6.436	221.591	212.06
5.637	38.253	10.099	194.626	169.78

Table 2.10: Added mass and Fluid damping results from pitch oscillation test, $Fn = 0.30$, zero trim and sinkage.

In the calculations of the added mass values the restoring coefficients of Table 2.6 are used, that is by zero forward speed, zero sinkage and zero trim angle. The results are also plotted in Figure 2.13 and 2.14.

$\omega\sqrt{L/g}$	A_{33} [kg]	A_{53} [kg m]	B_{33} [kg/s]	B_{53} [kg m/s]
3.217	45.575	37.070	217.715	16.008
3.748	48.026	33.691	236.840	5.971
4.157	48.725	30.963	268.449	14.973
4.392	49.224	29.834	293.516	22.439
4.582	48.268	28.354	313.831	32.303
4.898	47.154	25.933	346.016	45.200
5.286	44.734	22.617	389.196	59.611
5.741	42.358	19.085	426.478	67.402
6.277	40.425	15.783	469.392	68.117
6.961	39.818	13.595	514.114	54.944
7.890	40.316	12.299	593.556	38.714
7.921	40.545	12.477	599.563	41.143

Table 2.11: Added mass and Fluid damping results from heave oscillation test, $Fn = 0.45$, zero trim and sinkage

$\omega\sqrt{L/g}$	A_{55} [kg m ²]	A_{35} [kg m]	B_{55} [kg m ² /s]	B_{35} [kg m/s]
3.204	74.735	24.810	170.182	227.44
3.449	67.165	19.950	185.332	243.04
3.729	60.942	16.644	192.230	245.12
4.095	53.147	12.274	199.566	244.22
4.343	49.943	10.971	196.931	237.60
4.593	47.479	9.929	197.991	232.22
4.920	44.988	9.217	198.436	225.36
5.281	42.752	8.714	206.185	225.33
5.741	41.223	8.791	218.900	227.21
6.295	40.164	9.045	243.270	238.50
7.010	39.021	8.706	287.439	263.32

Table 2.12: Added mass and Fluid damping results from pitch oscillation test, $Fn = 0.45$, zero trim and sinkage.

In the calculations of the added mass values the restoring coefficients of Table 2.6 are used, that is by zero forward speed, zero sinkage and zero trim angle. The results are also plotted in Figure 2.15 and 2.16.

$\omega\sqrt{L/g}$	A_{33} [kg]	A_{53} [kg m]	B_{33} [kg/s]	B_{53} [kg m/s]
3.634	70.629	57.491	223.071	-28.230
4.304	61.628	41.561	257.578	-31.569
4.524	59.199	37.666	270.953	-32.881
4.712	57.556	34.589	280.368	-34.120
5.047	54.635	30.262	299.332	-36.500
5.408	52.622	26.315	323.823	-39.356
6.169	50.025	20.500	379.461	-47.356
7.484	48.468	15.005	502.204	-66.994

Table 2.13: Added mass and Fluid damping results from heave oscillation test, $Fn = 0.60$, zero trim and sinkage.

$\omega\sqrt{L/g}$	A_{55} [kg m ²]	A_{35} [kg m]	B_{55} [kg m ² /s]	B_{35} [kg m/s]
3.653	84.262	32.487	101.035	197.49
3.956	75.535	27.706	102.875	197.06
4.321	67.736	23.355	107.822	199.48
4.651	62.413	20.315	114.047	201.97
5.061	57.284	17.561	124.608	205.92
5.366	54.346	15.948	133.435	209.53
6.221	48.263	12.261	167.345	225.73
7.436	42.834	7.510	229.996	242.93

Table 2.14: Added mass and Fluid damping results from pitch oscillation test, $Fn = 0.60$, zero trim and sinkage

In the calculations of the added mass values the restoring coefficients of Table 2.6 are used, that is by zero forward speed, zero sinkage and zero trim angle. The results are also plotted in Figure 2.17 and 2.18.

$\omega\sqrt{L/g}$	A_{33} [kg]	A_{53} [kg m]	B_{33} [kg/s]	B_{53} [kg m/s]
4.103	58.127	36.576	259.164	-52.682
4.421	55.998	31.518	267.000	-60.045
4.861	54.010	26.244	295.647	-65.882
5.097	53.088	24.322	308.604	-69.784
5.439	52.701	21.716	326.846	-76.529
5.774	52.360	19.614	348.745	-84.486
6.120	52.838	18.201	374.428	-92.506
7.168	54.199	15.970	481.731	-109.533
8.662	55.925	15.337	721.833	-114.122

Table 2.15: Added mass and Fluid damping results from heave oscillation test, $Fn = 0.75$, zero trim and sinkage

$\omega\sqrt{L/g}$	A_{55} [kg m ²]	A_{35} [kg m]	B_{55} [kg m ² /s]	B_{35} [kg m/s]
4.094	68.276	15.143	115.602	270.08
4.880	58.724	12.224	140.366	293.85
4.464	62.987	13.461	125.966	281.70
5.126	56.374	11.240	152.147	309.13
5.423	54.252	10.306	166.420	316.93
5.766	52.161	8.981	184.783	326.59
6.138	50.059	7.471	207.337	340.21
7.158	45.929	4.397	278.008	369.68
8.687	41.741	1.460	405.697	388.30

Table 2.16: Added mass and Fluid damping results from pitch oscillation test, $Fn = 0.75$, zero trim and sinkage

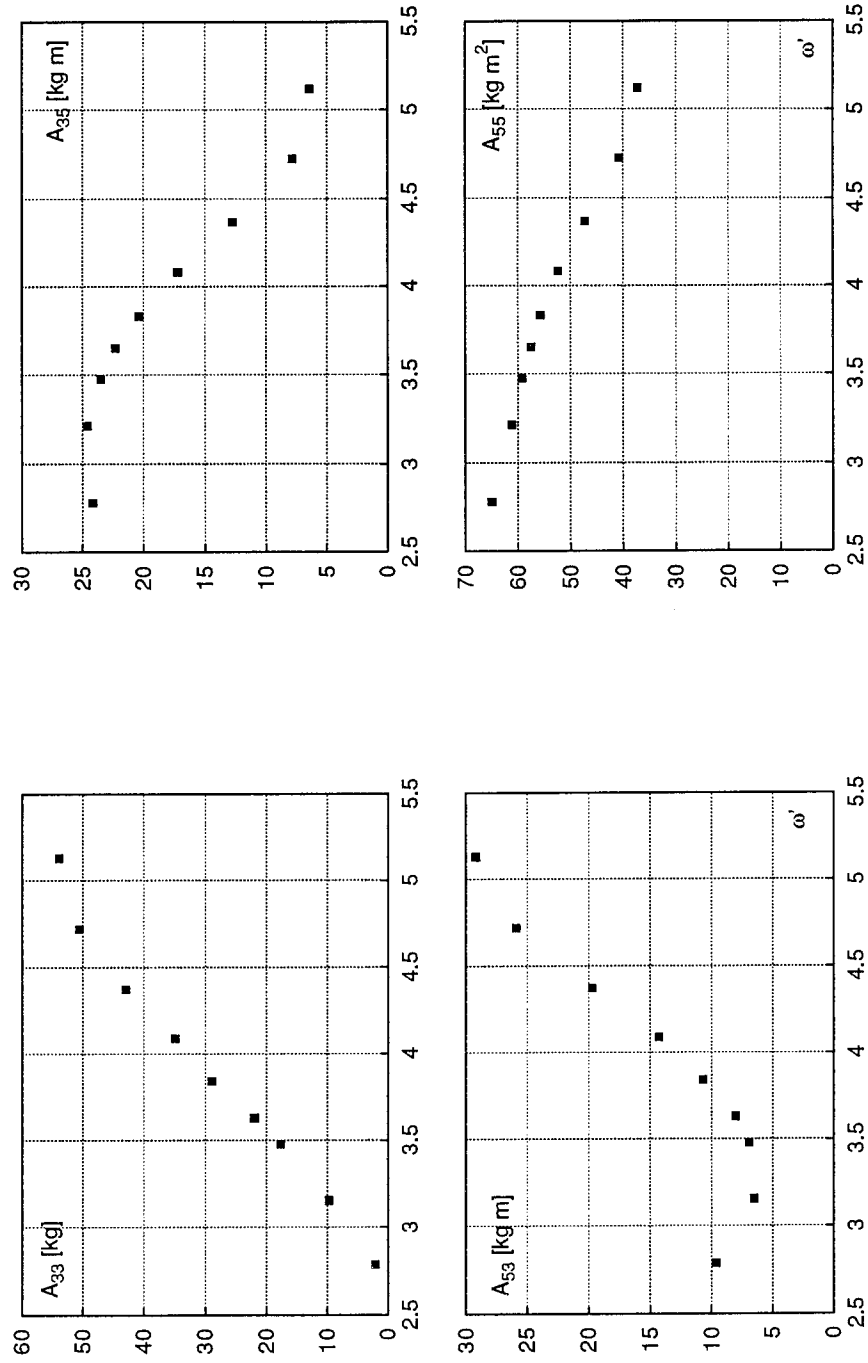


Figure 2.11: Added mass values, $Fn = 0.30$ ($\omega' = \omega\sqrt{L/g}$).

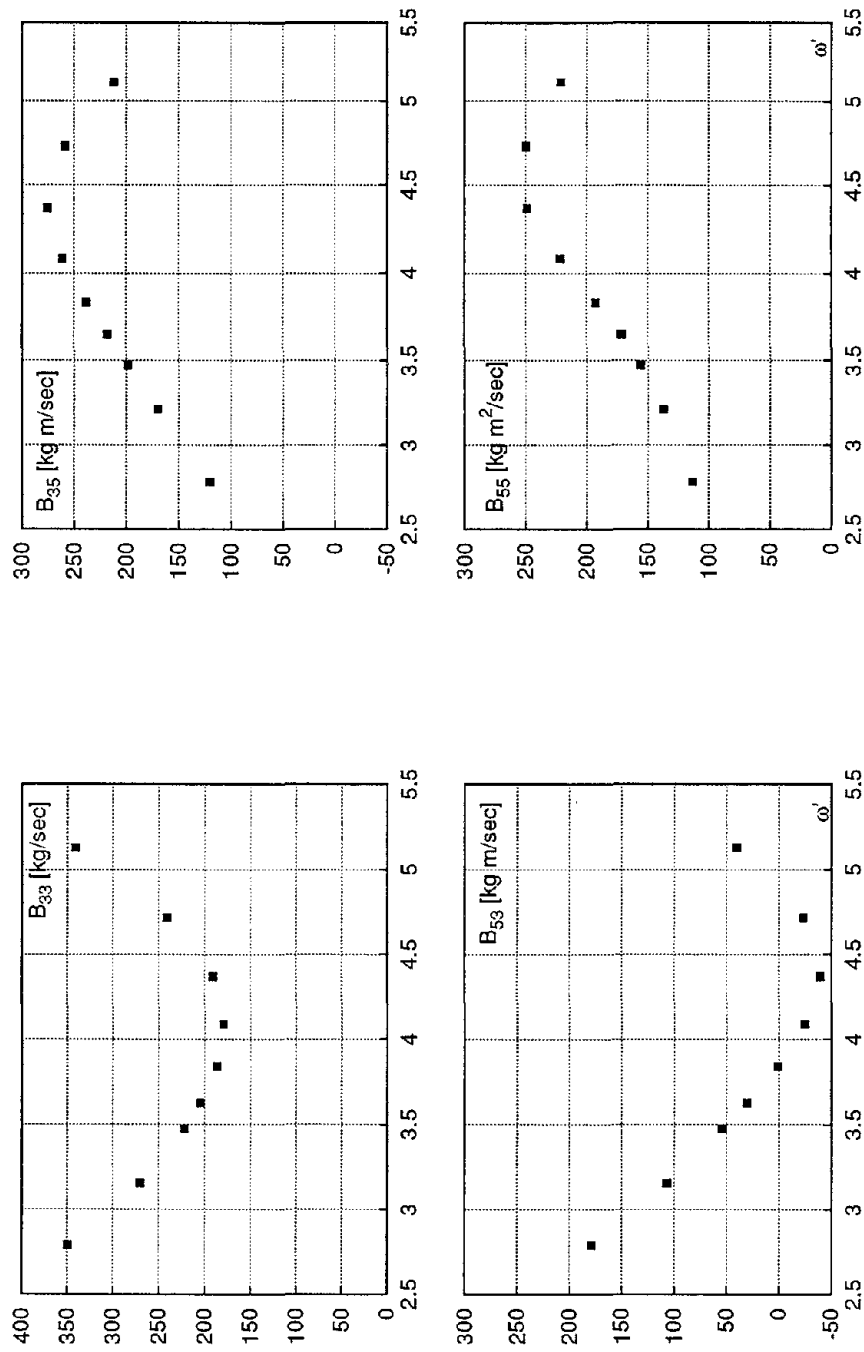


Figure 2.12: Fluid damping values, $Fn = 0.30$ ($\omega' = \omega \sqrt{L/g}$).

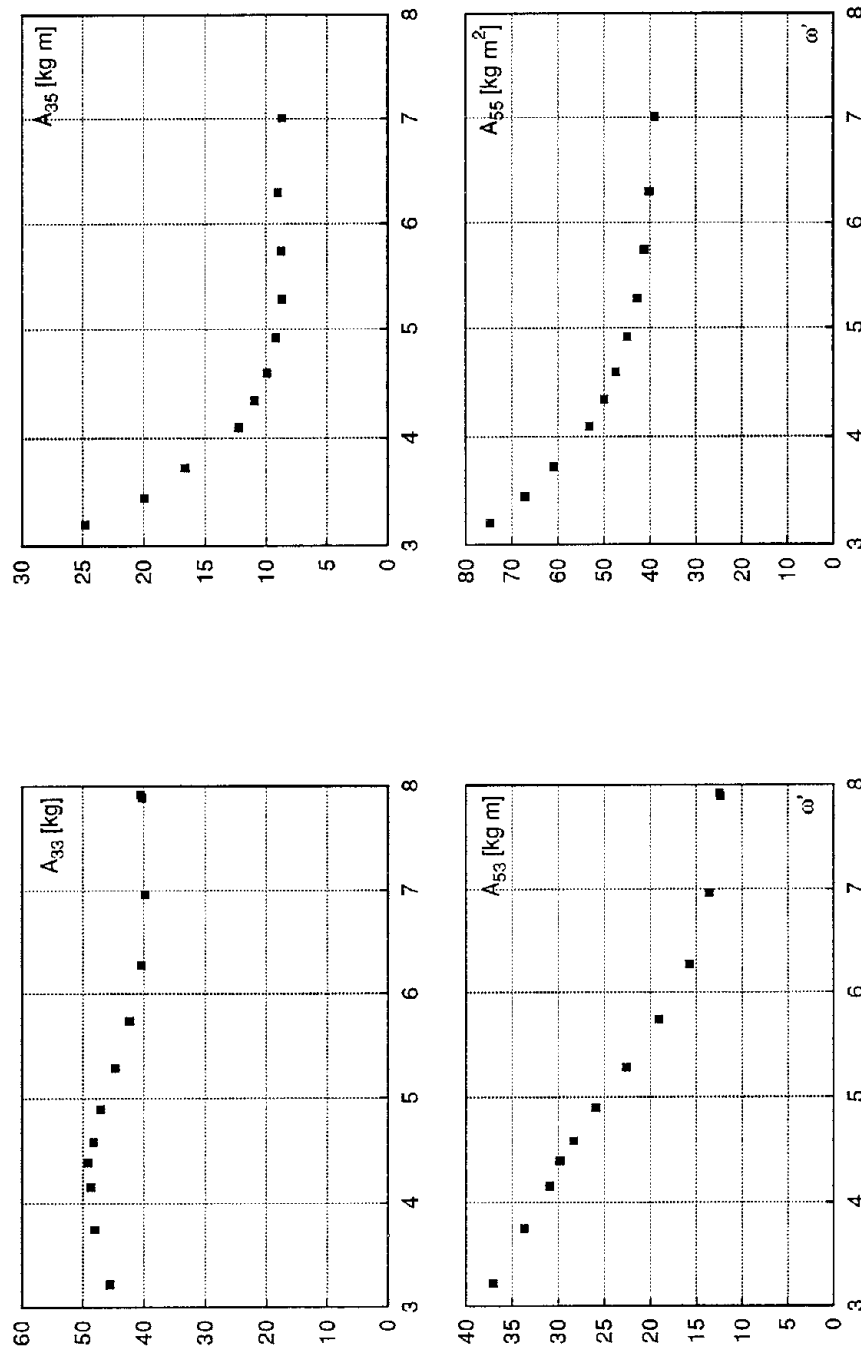


Figure 2.13: Added mass values, $Fn = 0.45$ ($\omega' = \omega \sqrt{L/g}$).

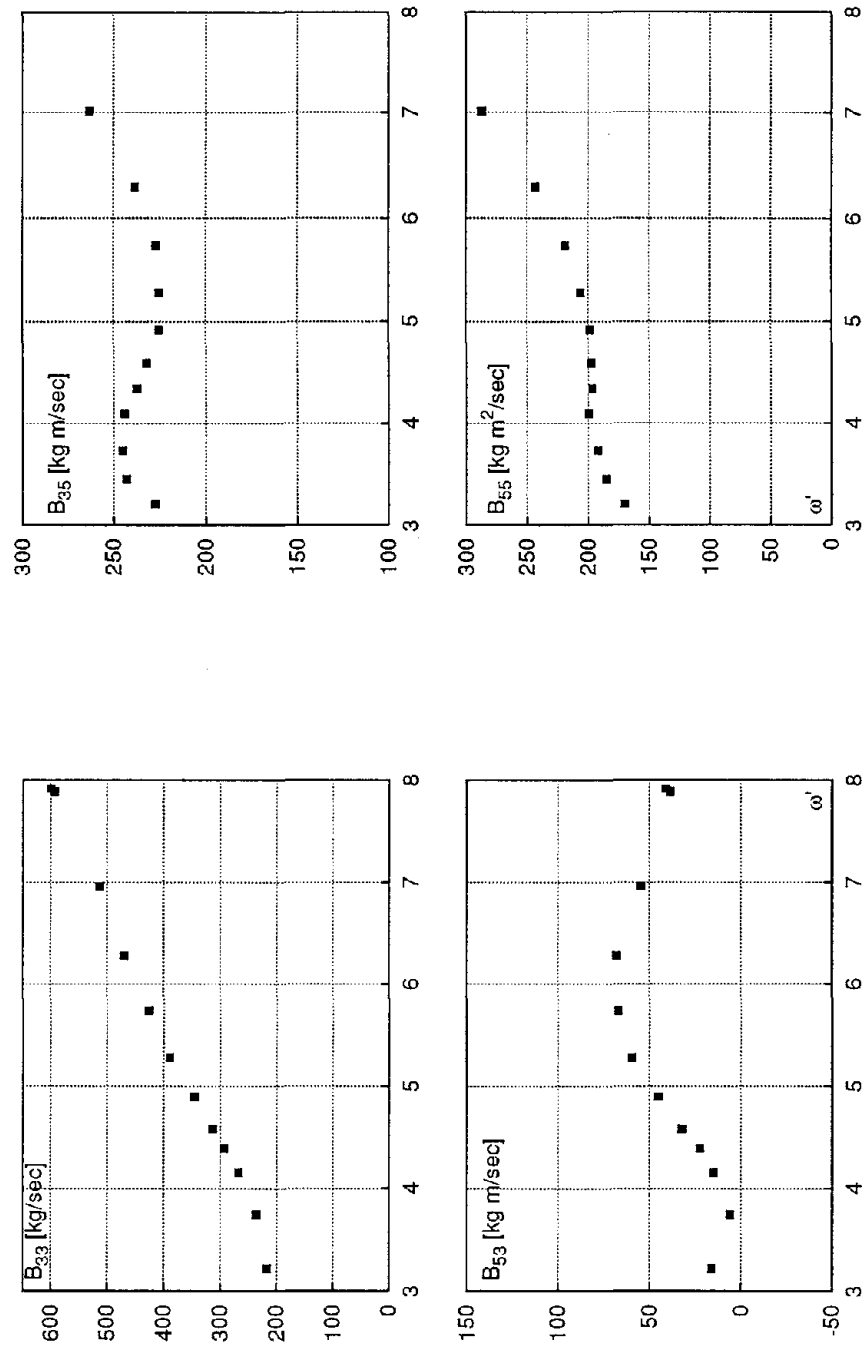


Figure 2.14: Fluid damping values, $Fn = 0.45$ ($\omega' = \omega\sqrt{L/g}$).

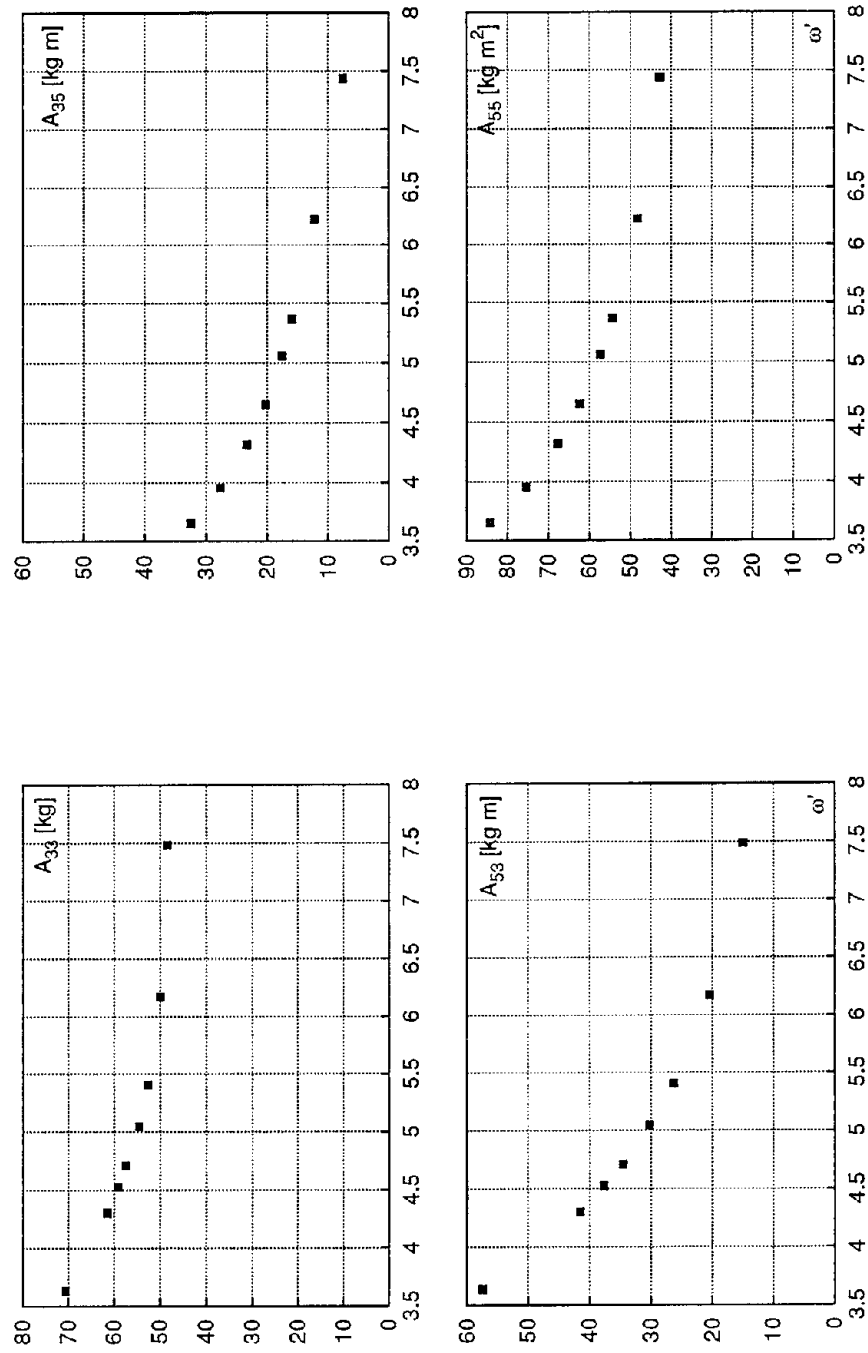


Figure 2.15: Added mass values, $Fn = 0.60$ ($\omega' = \omega\sqrt{L/g}$).

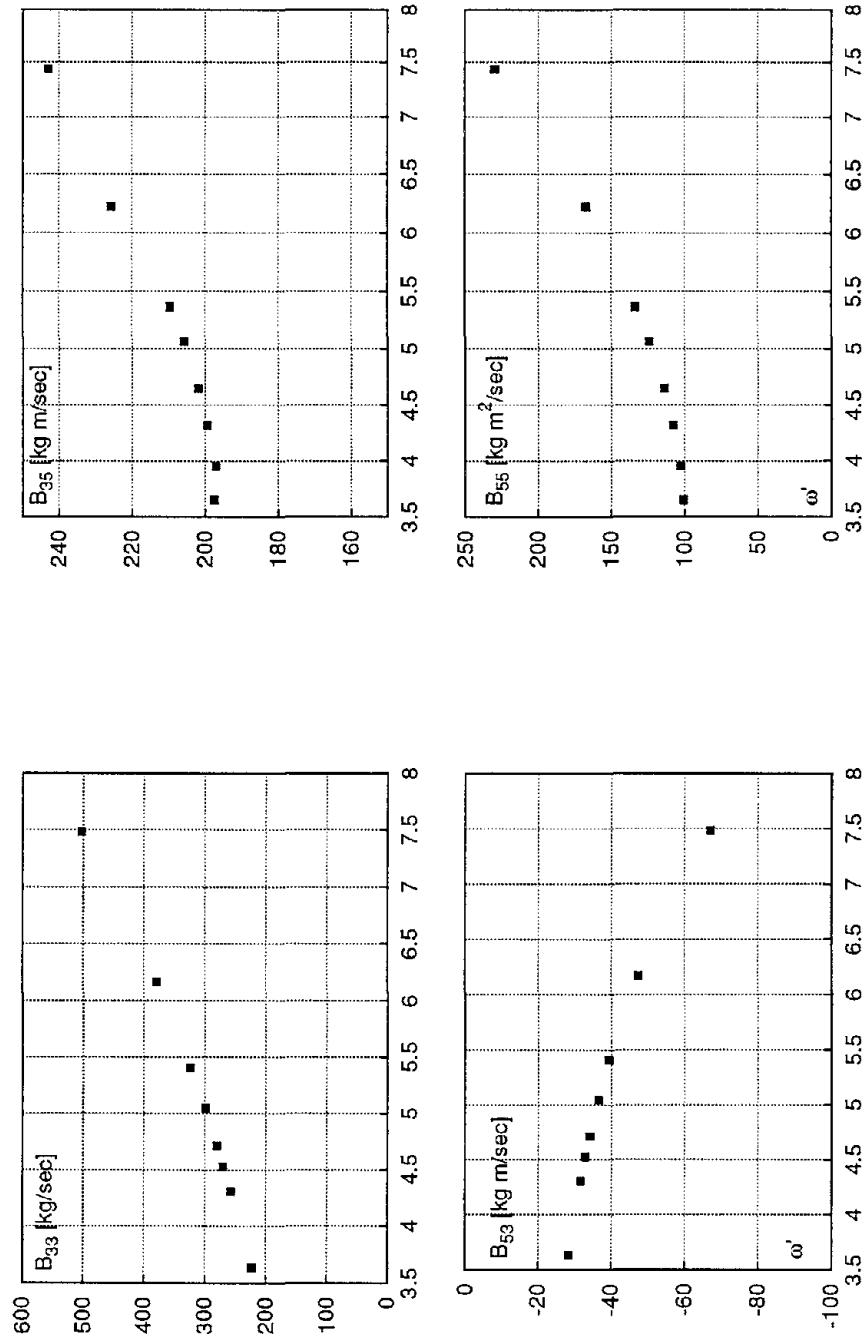


Figure 2.16: Fluid damping values, $Fn = 0.60$ ($\omega' = \omega\sqrt{L/g}$).

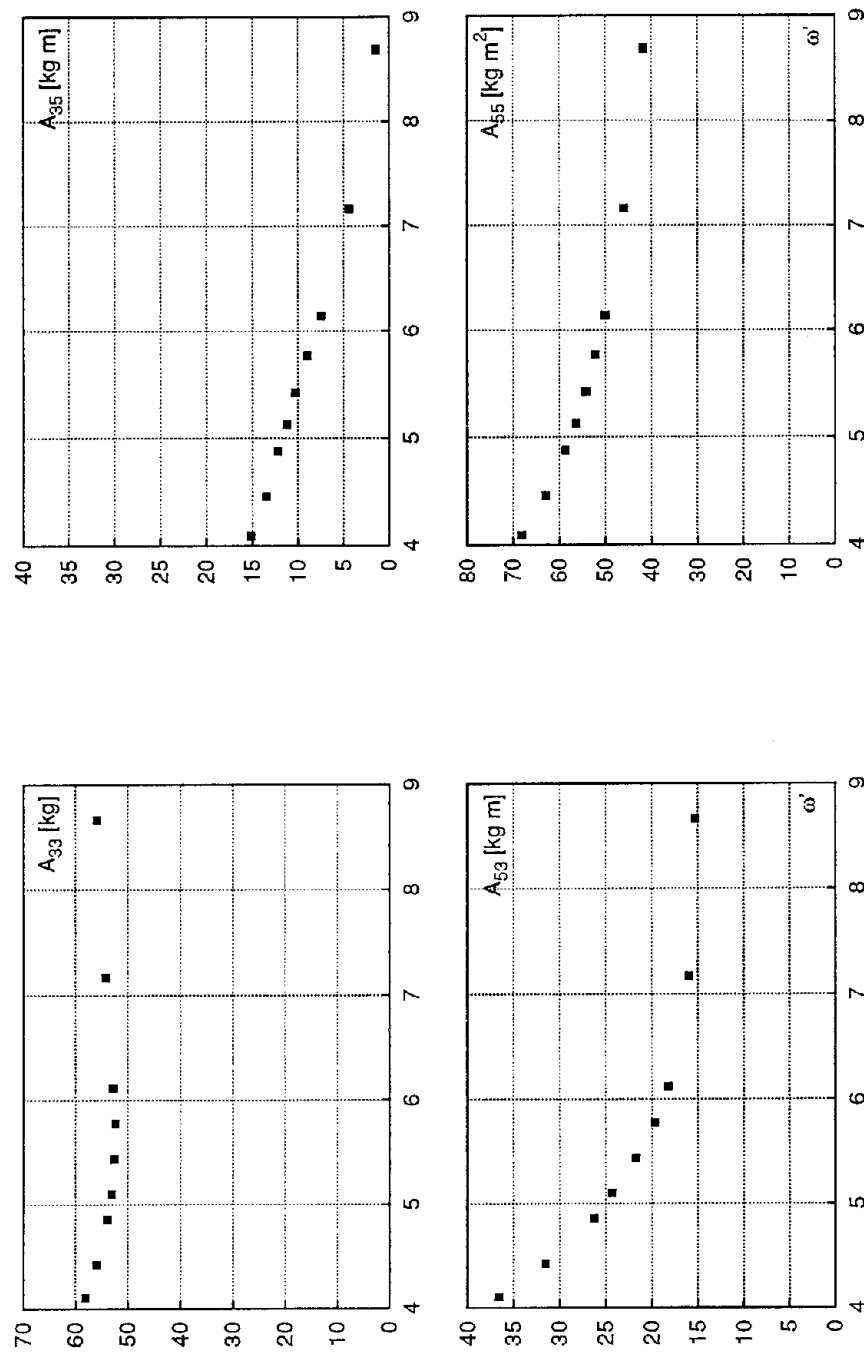


Figure 2.17: Added mass values, $Fn = 0.75$ ($\omega' = \omega\sqrt{L/g}$).

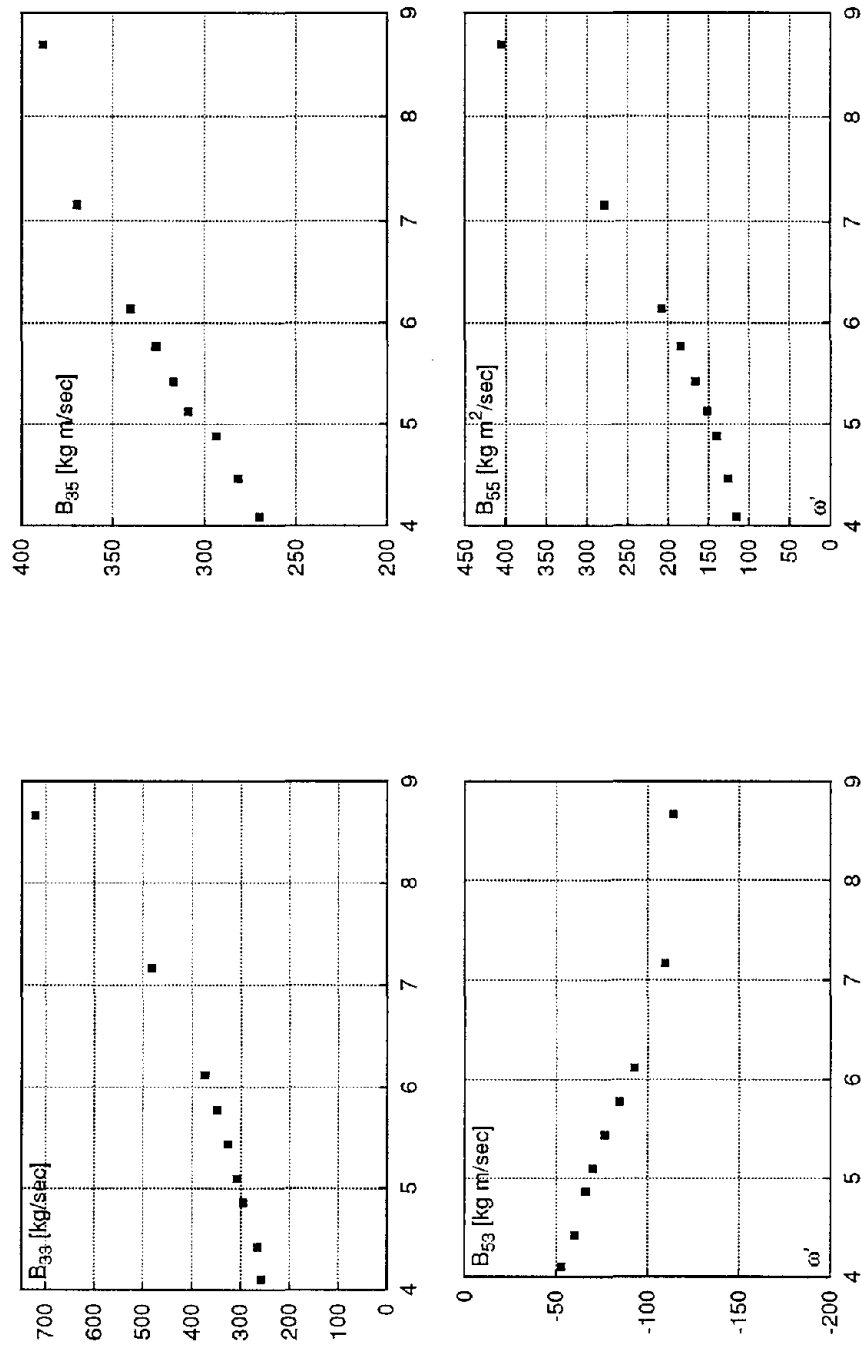


Figure 2.18: Fluid damping values, $Fn = 0.75$ ($\omega' = \omega\sqrt{L/g}$).

In the calculations of the added mass values the restoring coefficients of Table 2.6 are used, that is by zero forward speed, zero sinkage and zero trim angle. The results are also plotted in Figure 2.19 and 2.20. Due to experimental limitations the non dimensional frequency ($\omega\sqrt{L/g}$) could not exceed 6.2 during the pitch oscillation tests.

$\omega\sqrt{L/g}$	A ₃₃ [kg]	A ₅₃ [kg m]	B ₃₃ [kg/s]	B ₅₃ [kg m/s]
4.123	59.039	20.084	375.948	-39.682
5.455	62.869	17.779	338.172	-66.809
7.140	69.966	19.265	303.824	-100.163
6.146	64.609	17.675	317.634	-81.610
4.885	61.718	18.402	358.924	-51.615
4.467	60.203	19.202	377.239	-40.175
8.672	74.236	20.212	320.303	-103.799

Table 2.17: Added mass and Fluid damping results from heave oscillation test, $Fn = 0.75$, trim and sinkage correction.

$\omega\sqrt{L/g}$	A ₅₅ [kg m ²]	A ₃₅ [kg m]	B ₅₅ [kg m ² /s]	B ₃₅ [kg m/s]
4.138	63.847	8.376	255.602	413.52
4.467	61.309	10.774	255.014	418.05
4.877	58.933	12.703	256.807	430.26
5.440	56.269	14.337	262.963	447.97
6.139	54.308	15.355	271.316	469.76

Table 2.18: Added mass and Fluid damping results from pitch oscillation test, $Fn = 0.75$, trim and sinkage correction.

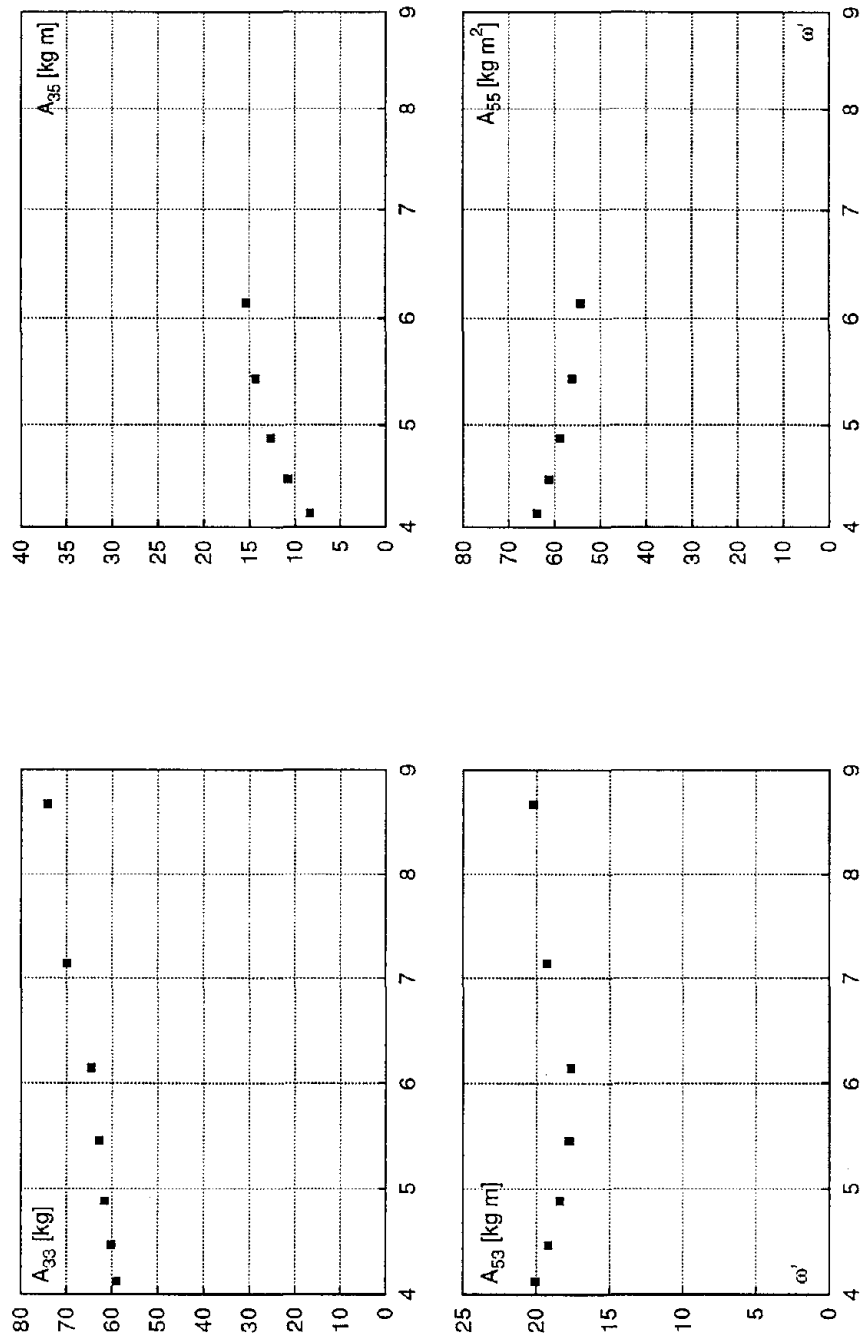


Figure 2.19: Added mass values, $Fn = 0.75$ ($\omega' = \omega\sqrt{L/g}$).

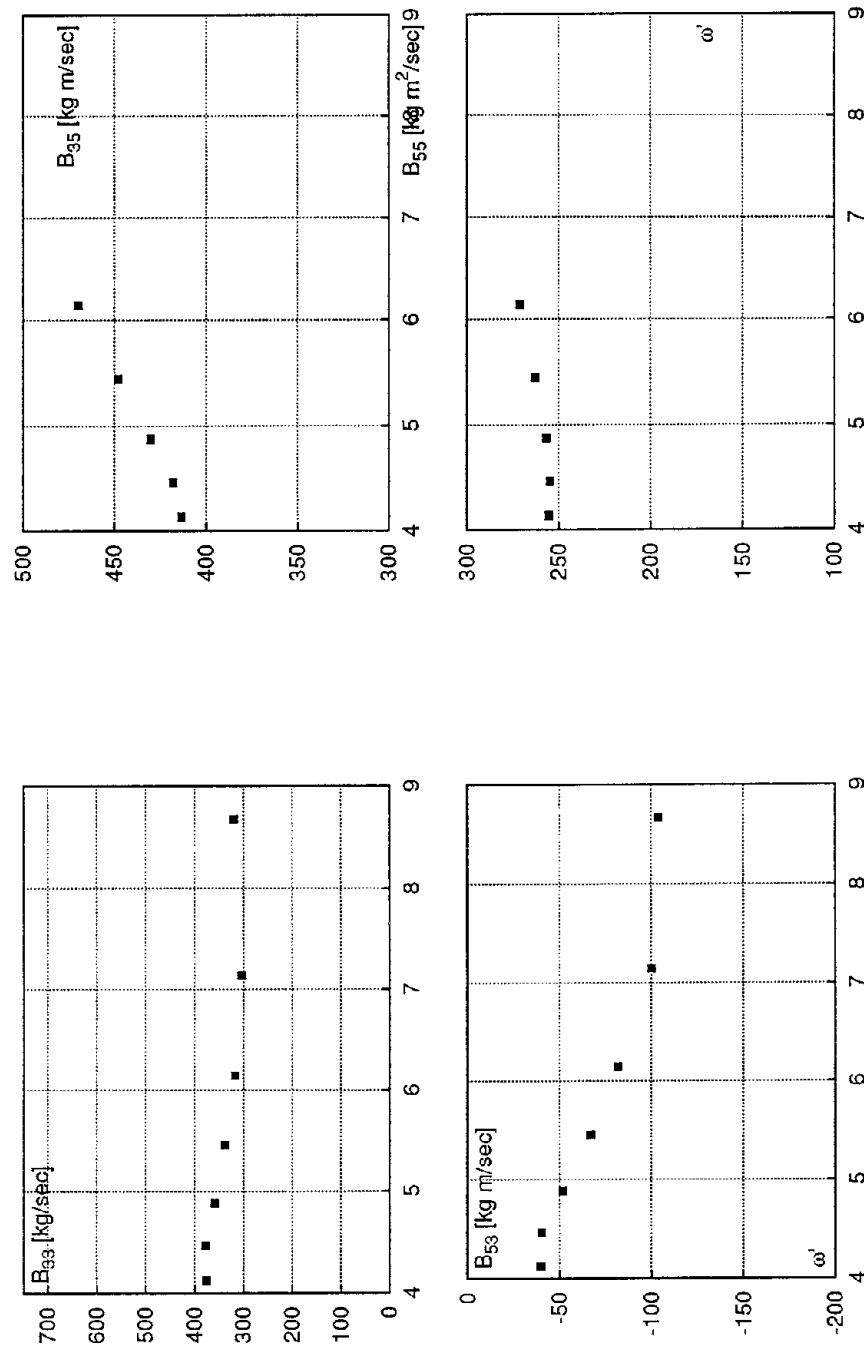


Figure 2.20: Fluid damping values, $Fn = 0.75$ ($\omega' = \omega \sqrt{L/g}$).

2.4 The wave load tests

Table:

2.19: Wave force F_z and M_y , zero trim and sinkage, $Fn = 0.30$

2.20: Wave force F_z and M_y , zero trim and sinkage, $Fn = 0.45$

2.21: Wave force F_z and M_y , zero trim and sinkage, $Fn = 0.60$

2.22: Wave force F_z and M_y , zero trim and sinkage, $Fn = 0.75$

Figure:

2.21: Wave force F_z and M_y , zero trim and sinkage, $Fn = 0.30$

2.22: Wave force F_z and M_y , zero trim and sinkage, $Fn = 0.45$

2.23: Wave force F_z and M_y , zero trim and sinkage, $Fn = 0.60$

2.24: Wave force F_z and M_y , zero trim and sinkage, $Fn = 0.75$

The vertical wave force and the wave moment around the y-axis are presented in the tables and figures below. During the experiments the vessel was kept fixed on its zero forward speed waterline. It is understood that this is not the actual position of the vessel if the vessel has a certain forward speed since then a trim and sinkage correction should be taken into account.

λ/L	ζ_a [mm]	F_{3a} [N]	ϵ_3 [deg]	F_{5a} [Nm]	ϵ_5 [deg]
0.604	22.620	38.627	-164.473	25.996	-137.288
0.685	21.232	26.497	-138.098	41.919	-97.903
0.804	20.571	17.194	-68.695	63.022	-86.275
1.000	18.338	48.763	-8.748	80.002	-74.622
1.098	18.139	63.958	-1.388	82.566	-71.610
1.199	18.003	77.256	2.344	83.613	-69.813
1.299	18.075	89.946	5.042	83.739	-68.221
1.399	18.012	101.523	6.346	82.281	-66.972
1.597	18.311	122.045	7.805	81.188	-65.530
1.989	19.366	150.747	9.029	72.178	-63.285

Table 2.19: Wave loads in head waves, F_3 and F_5 , $Fn = 0.30$.

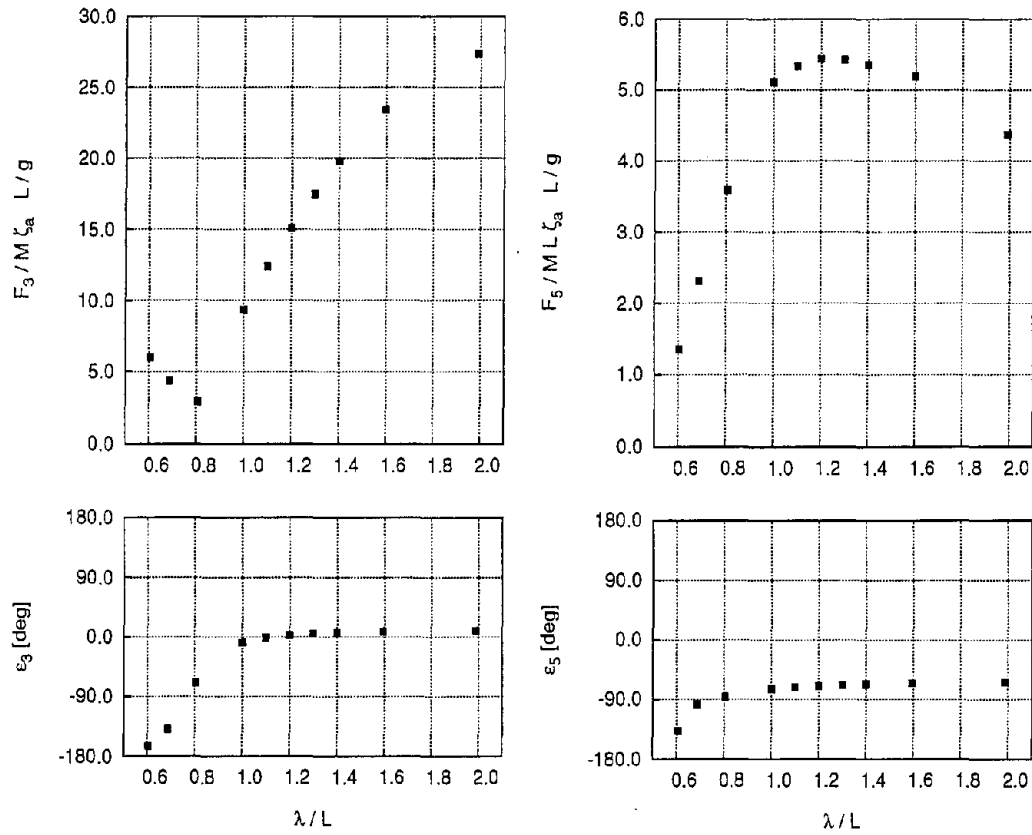


Figure 2.21: Wave loads in head waves, F_3 and F_5 , $Fn = 0.30$.

λ/L	ζ_a [mm]	F_{3a} [N]	ϵ_3 [deg]	F_{5a} [Nm]	ϵ_5 [deg]
0.602	23.005	24.790	77.343	15.529	97.206
0.602	22.295	23.727	-154.561	15.293	-134.000
0.685	22.210	19.321	-133.977	30.989	-94.724
0.798	19.999	14.011	-73.340	51.614	-83.747
1.001	19.082	40.220	-12.350	73.416	-78.121
1.109	18.034	54.208	-5.512	75.783	-75.954
1.195	18.408	68.485	-3.440	80.542	-75.622
1.296	18.658	81.575	1.070	81.709	-72.763
1.399	18.799	92.110	3.131	80.267	-71.542
1.593	19.158	112.625	5.789	79.689	-69.103
1.973	21.014	141.908	9.480	72.581	-65.764

Table 2.20: Wave loads in head waves, F_3 and F_5 , $Fn = 0.45$.

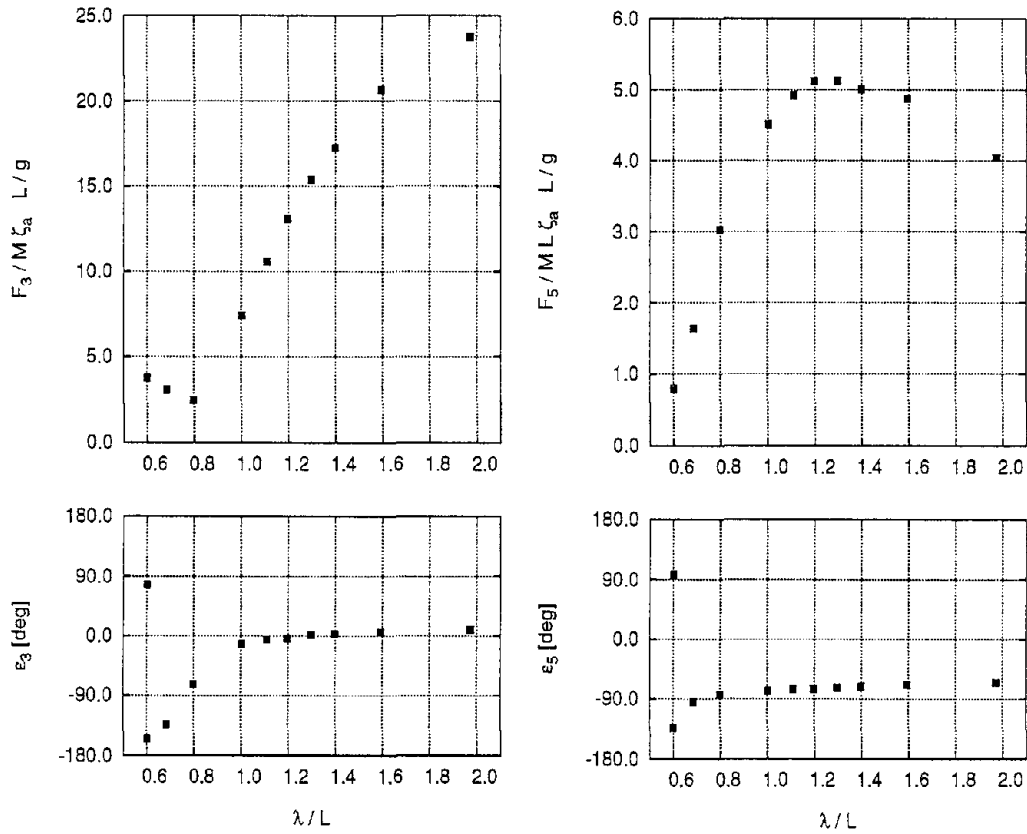


Figure 2.22: Wave loads in head waves, F_3 and F_5 , $Fn = 0.45$.

λ/L	ζ_a [mm]	F_{3a} [N]	ϵ_3 [deg]	F_{5a} [Nm]	ϵ_5 [deg]
0.800	29.547	12.149	-21.456	42.028	-66.706
0.908	19.612	25.051	6.132	50.251	-65.175
1.002	19.261	37.743	9.635	56.552	-68.945
1.197	18.921	60.602	12.242	61.088	-72.347
1.297	18.744	71.152	13.825	62.094	-72.497
1.396	18.889	80.476	15.177	62.079	-72.760
1.600	18.897	96.553	13.068	61.064	-76.666
1.769	19.205	110.819	11.072	60.220	-79.893
1.980	19.935	123.557	10.412	58.482	-82.063

Table 2.21: Wave loads in head waves, F_3 and F_5 , $Fn = 0.60$.

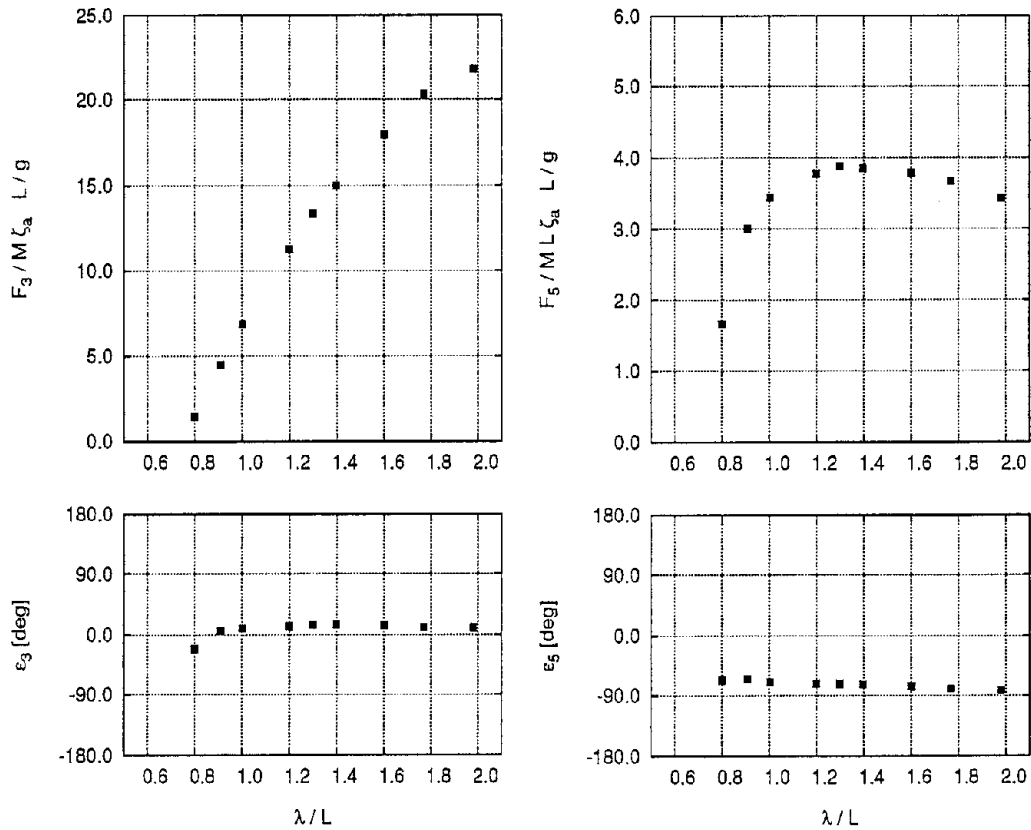


Figure 2.23: Wave loads in head waves, F_3 and F_5 , $Fn = 0.60$.

λ/L	ζ_a [mm]	F_{3a} [N]	ϵ_3 [deg]	F_{5a} [Nm]	ϵ_5 [deg]
0.800	19.699	14.522	-77.594	35.970	-74.990
0.899	19.199	19.209	-23.570	49.764	-72.049
0.998	21.217	31.342	-4.724	58.794	-71.741
1.097	18.811	42.516	2.126	63.596	-72.514
1.202	17.730	54.846	7.496	66.028	-72.257
1.295	17.986	66.211	8.128	68.615	-72.807
1.401	17.417	75.217	8.787	69.234	-74.283
1.599	17.088	95.342	10.953	69.412	-74.637
1.986	18.359	121.781	10.404	66.802	-77.595

Table 2.22: Wave loads in head waves, F_3 and F_5 , $Fn = 0.75$.

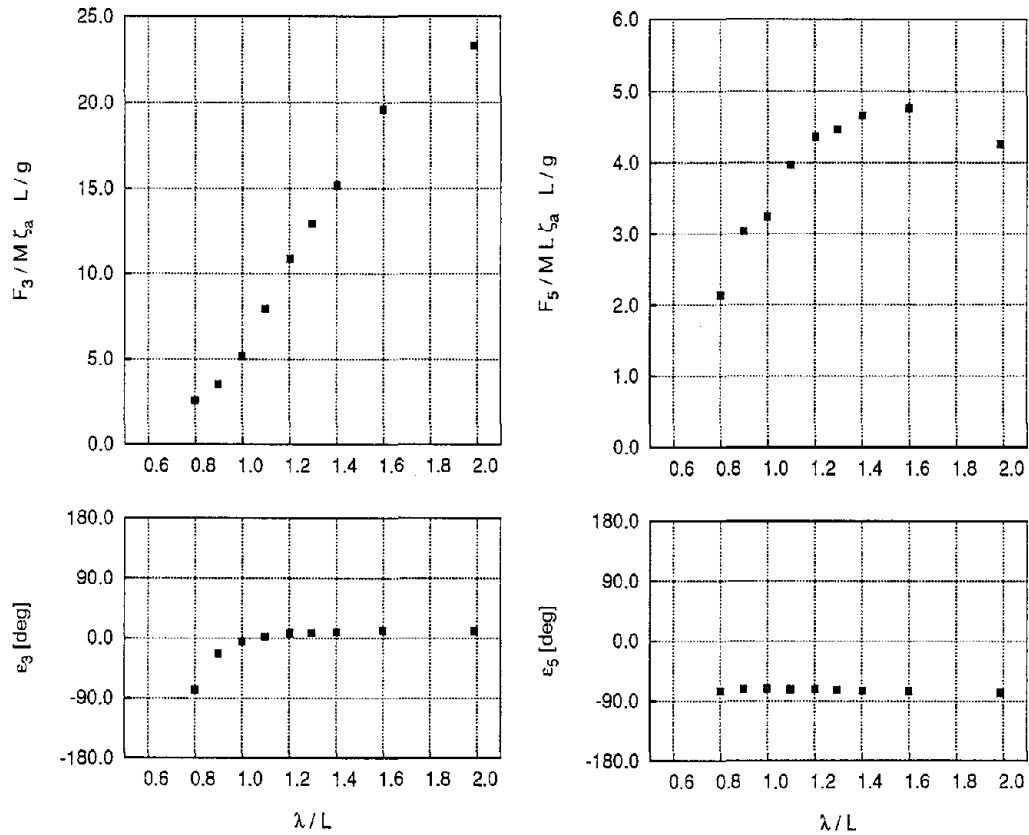


Figure 2.24: Wave loads in head waves, F_3 and F_5 , $Fn = 0.75$.

Acknowledgment

The catamaran hull form has been designed by Aad Versluis and has been manufactured at the Delft Shiphydromechanic Laboratory by Cees van den Bergh. The mechanical layout and construction parts for the model has been designed by Peter Poot and build by Hans van der Hek.

The electronic set-up has been realised by Rein van den Oever, who also performed most of the measurements in corporation with the author. His contribution to the project is especially appreciated.

Bibliography

Van 't Veer, A. P.: 1998a, *Behaviour of catamaran vessels in head and oblique waves*, PhD thesis, Delft University of Technology.

Van 't Veer, A. P.: 1998b, Experimental results of motions and structural loads on the 372 catamaran model in head and oblique waves, *Technical Report 1130*, Delft University of Technology.